

UNIT 1: MATHEMATICAL OPERATIONS ON WHOLE NUMBERS UP TO 100,000

1. INTRODUCTION

Numbers are very important in our daily life.

We use numbers to:

- Count things.
- Add, subtract, multiply and divide.
- Count money.
- Buy and sell goods.
- Pay school fees.
- Tell time.
- Dial telephone numbers.
- Solve everyday problems.

2. PLACE VALUE OF WHOLE NUMBERS UP TO 100,000

Place value is the value of a digit according to its position in a number.

The main place values are:

Ten Thousands (T.Th)

Thousands (Th)

Hundreds (H)

Tens (T)

Ones (O)

Example:

Number: 34,271

Digit 3 → Ten Thousands → 30,000

Digit 4 → Thousands → 4,000

Digit 2 → Hundreds → 200

Digit 7 → Tens → 70

Digit 1 → Ones → 1

Therefore:

$$34,271 = 30,000 + 4,000 + 200 + 70 + 1$$

REMEMBER:

The position of a digit determines its value.

EXERCISE 1

Give the place value of the indicated digit.

1. What is the place value of 8 in 18,245?
2. What is the place value of 7 in 47,321?
3. What is the place value of 6 in 3,674?
4. What is the place value of 9 in 29,321?
5. What is the value of 4 in 34,271?

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3. EXPANDING WHOLE NUMBERS

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Expanded form means breaking a number into the values of its digits.

Example 1:

82,645

$$= 80,000 + 2,000 + 600 + 40 + 5$$

Example 2:

34,271

$$= 30,000 + 4,000 + 200 + 70 + 1$$

SHORT FORM

We can also change expanded form back into a number.

Example:

$$30,000 + 400 + 40 + 4$$

$$= 30,444$$

EXERCISE 2

A. Write these numbers in expanded form.

1. 4,624
2. 984
3. 44,484
4. 93,428
5. 76,709

B. Write these in short form.

1. $7,000 + 300 + 80 + 3$
2. $90,000 + 8,000 + 300 + 70 + 6$
3. $30,000 + 9,000 + 800 + 60 + 7$
4. $50,000 + 4,000 + 200 + 10 + 9$

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4. FORMING NUMBERS UP TO 100,000

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Digits are the numbers from 0 to 9.

They are:

0, 1, 2, 3, 4, 5, 6, 7, 8 and 9.

We arrange digits in different positions to form different numbers.

Example:

Using the digits:

4, 6, 7, 0 and 1

we can form:

46,701
70,146
64,701
14,670
10,467

Changing the position of digits changes the number.

Example:

Using 1, 3, 5, 9 and 4:

Smallest 5-digit number = 13,459

Largest 5-digit number = 95,431

EXERCISE 3

Use the digits 5, 2, 4, 1 and 3 once each.

1. Form a number beginning with 5.
2. Form a number beginning with 2.
3. Form a number beginning with 4.
4. Form a number beginning with 1.
5. Form the largest possible number.
6. Form the smallest possible number.

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5. READING AND WRITING NUMBERS UP TO 100,000

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We can write numbers in:

- Figures
- Words

Examples:

10 = Ten

30 = Thirty

100 = One hundred

370 = Three hundred seventy

1,000 = One thousand

8,900 = Eight thousand nine hundred

50,000 = Fifty thousand

99,990 = Ninety-nine thousand, nine hundred ninety

100,000 = One hundred thousand

Example:

48,625

= Forty-eight thousand, six hundred twenty-five.

92,100

= Ninety-two thousand, one hundred.

EXERCISE 4

A. Write these numbers in words.

1. 111

2. 1,011

3. 54,700

4. 90,001

5. 78,999

6. 15,000

7. 10,002

8. 97,642

B. Write these in figures.

1. Ninety-nine

2. One hundred ninety-nine

3. Seven hundred five

4. Six hundred fifty

5. Twelve thousand, five hundred

6. Eighty thousand

7. Forty thousand one

8. One hundred thousand

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6. COMPARING WHOLE NUMBERS

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We use these signs:

< means "less than"

> means "greater than"

= means "equal to"

RULE 1:

If two numbers have different numbers of digits, the number with more digits is greater.

Example:

10,000 > 1,000

RULE 2:

If the numbers have the same number of digits, compare their digits starting from the left.

Example:

50,000 and 55,000

Both have five digits.

Compare from the left:

5 = 5

Next:

0 < 5

Therefore:

50,000 < 55,000

EXERCISE 5

Put <, > or =.

1. 2,000 _____ 20,000

2. 99,999 _____ 9,999

3. 70,001 _____ 71,000

4. 45,000 _____ 45,000

5. 23,456 _____ 23,465

6. 89,000 _____ 88,999

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7. ASCENDING AND DESCENDING ORDER

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ASCENDING ORDER

Ascending order means arranging numbers from the smallest to the largest.

Example:

56,789; 12,345; 34,567; 90,123

Ascending order:

12,345; 34,567; 56,789; 90,123

DESCENDING ORDER

Descending order means arranging numbers from the largest to the smallest.

Example:

78,901; 45,678; 99,999; 12,345

Descending order:

99,999; 78,901; 45,678; 12,345

REMEMBER:

Ascending = Smallest → Largest

Descending = Largest → Smallest

EXERCISE 6

1. Arrange in ascending order:

23,456; 12,345; 56,789; 34,567

2. Arrange in descending order:

78,901; 65,432; 89,012; 54,321

3. Arrange in ascending order:

50,000; 35,000; 72,000; 15,000

4. Arrange in descending order:

99,999; 45,678; 87,654; 12,345

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8. ADDITION OF WHOLE NUMBERS

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Addition means putting numbers together to find their total.

When adding numbers:

- Ones must be under ones.
- Tens must be under tens.
- Hundreds must be under hundreds.
- Thousands must be under thousands.
- Ten thousands must be under ten thousands.

Start adding from the ones column.

A. ADDITION WITHOUT CARRYING

Example:

$$\begin{array}{r} 43,534 \\ +21,153 \\ \hline 64,687 \end{array}$$

B. ADDITION WITH CARRYING

Sometimes the total in a column is greater than 9.

Write the ones digit and carry the tens digit to the next column.

Example:

$$4,999 + 8,294 = 13,293$$

EXERCISE 7

Calculate:

1. $53,250 + 32,488$

2. $738 + 262$

3. $1,354 + 9,999$

4. $999 + 111$

5. $999 + 1,110$

WORD PROBLEMS

6. One school received 42,130 textbooks and another school received 32,041 textbooks.

How many textbooks did the two schools receive altogether?

7. Jean delivered 58,746 kg of coffee and Aline delivered 46,978 kg.

Find the total amount of coffee delivered.

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9. SUBTRACTION OF WHOLE NUMBERS

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Subtraction means taking one quantity away from another or finding the difference between two numbers.

A. SUBTRACTION WITHOUT BORROWING

Arrange numbers according to their place values.

Start subtracting from the ones.

Example:

$$6,989 - 3,453 = 3,536$$

B. SUBTRACTION WITH BORROWING

Sometimes the upper digit is smaller than the digit below it.

We borrow from the next place value.

Borrowing is also called regrouping.

Example:

$$2,573 - 1,395$$

Borrow where necessary.

Answer:

$$2,573 - 1,395 = 1,178$$

EXERCISE 8

Calculate:

1. $15,789 - 11,000$

2. $86,786 - 63,524$

3. $99,999 - 29,999$

4. $32,450 - 16,360$

5. $54,000 - 543$

6. $65,009 - 222$

7. $1,000 - 1$

WORD PROBLEMS

8. A shirt costs 3,000 Frw and another shirt costs 1,999 Frw.

Find the difference in their prices.

9. A district had 30,845 people. Of these, 19,678 were females.

How many were males?

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10. MULTIPLICATION OF WHOLE NUMBERS

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Multiplication is a quick way of adding equal groups.

Example:

$$9 \times 7 = 63$$

Learners should practise multiplication tables.

A. QUICK MULTIPLICATION

Multiplying by 10:

Add one zero.

$$237 \times 10 = 2,370$$

Multiplying by 100:

Add two zeroes.

$$457 \times 100 = 45,700$$

Multiplying by 1,000:

Add three zeroes.

$$34 \times 1,000 = 34,000$$

Multiplying by 10,000:

Add four zeroes.

$$6 \times 10,000 = 60,000$$

MULTIPLYING BY 5

A quick method is:

1. Multiply by 10.
2. Divide the answer by 2.

Example:

$$280 \times 5$$

$$= (280 \times 10) \div 2$$

$$= 2,800 \div 2$$

$$= 1,400$$

EXERCISE 9

Calculate:

1. 164×6

2. 40×9

3. 78×7

4. 217×5

5. 450×5

6. 897×5

7. 575×3

8. 349×8

Quick multiplication:

9. $2 \times 10,000$

10. $95 \times 1,000$

11. 899×100

12. $9,000 \times 10$

13. 120×5

14. 450×5

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11. MULTIPLYING BY A TWO-DIGIT NUMBER

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We can multiply a whole number by a two-digit number to find large totals quickly.

Example:

$$26 \times 26 = 676$$

EXERCISE 10

1. Find:

$$21 \times 13$$

2. Find:

$$476 \times 25$$

3. A rectangular room has 26 tiles along its length and 15 tiles along its width.

How many tiles cover the floor?

4. One notebook costs 450 Frw.

Find the cost of 35 notebooks.

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12. DIVISION OF WHOLE NUMBERS

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Division means splitting a number into equal parts.

A. DIVISION WITHOUT A REMAINDER

When nothing remains after division, the division is exact.

Examples:

$$24 \div 6 = 4$$

$$150 \div 5 = 30$$

EXERCISE 11

Calculate:

1. $840 \div 10$

2. $180 \div 4$

3. $330 \div 11$

4. $2,247 \div 7$

5. There are 340 bags of cement to be unloaded equally by 5 men.

How many bags will each man unload?

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13. DIVISION WITH A REMAINDER

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Sometimes a number cannot be divided exactly.

The amount left over is called the REMAINDER.

Example:

$$25 \div 4$$

$$4 \times 6 = 24$$

$$25 - 24 = 1$$

Therefore:

$$25 \div 4 = 6 \text{ remainder } 1$$

Another example:

$$17 \div 5 = 3 \text{ remainder } 2$$

EXERCISE 12

1. $29 \div 4$

2. $38 \div 5$

3. $47 \div 6$

4. $65 \div 8$

5. A class has 67 pupils. The teacher puts them into groups of 4.

a) How many complete groups can be formed?

b) How many pupils remain?

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UNIT 1 REVISION EXERCISE

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1. Using the digits 1, 2, 3, 4 and 5:

a) Form the largest possible 5-digit number.

b) Form the smallest possible 5-digit number.

2. Write in figures:

a) Twenty-four thousand seven hundred seven.

b) One hundred thousand.

c) Thirty-four thousand seven hundred seventy-seven.

3. Write 67,405 in expanded form.

4. What is the value of 8 in 38,526?

5. Put $<$, $>$ or $=$:

45,678 _____ 45,687

6. Arrange in ascending order:

45,000; 12,000; 67,000; 23,000.

7. A shopkeeper's sales were:

Monday = 24,750 Frw

Tuesday = 24,450 Frw

Wednesday = 24,900 Frw

How much money did the shopkeeper receive altogether?

8. A water tank contains 100,000 litres.

P4 pupils use 12,500 litres.

How many litres remain?

9. Each tray contains 30 eggs.

How many eggs are there in 22 trays?

10. A father has 23 sweets and shares them equally among four children.

a) How many sweets does each child receive?

b) How many sweets remain?

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QUICK REVISION

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- Place value tells us the value of a digit according to its position.
- Expanded form breaks a number into its place values.
- < means less than.
- > means greater than.
- = means equal to.

- Ascending order = smallest to largest.
- Descending order = largest to smallest.
- Addition finds the total.
- Subtraction finds the difference or amount remaining.
- Multiplication is repeated addition.
- Division separates a quantity into equal groups.
- A remainder is what is left after a number cannot be divided exactly.
- Always arrange digits according to their correct place values when adding or subtracting.

END OF UNIT 1

UNIT 2: POSITIVE AND NEGATIVE INTEGERS

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1. INTRODUCTION TO INTEGERS

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Integers are used in many situations in everyday life.

Examples:

- Temperature:
+30°C can show a hot temperature.
-5°C can show a cold temperature.
- Bank balances:
Positive numbers can show money available or deposited.
Negative numbers can show money withdrawn or owed.
- Games:
Positive numbers can show points gained.
Negative numbers can show points lost.
- Height and depth:
A position above the ground can be positive.
A position below the ground can be negative.

Example:

3 metres above the ground = +3 m

2 metres below the ground = -2 m

2. MEANING OF POSITIVE AND NEGATIVE NUMBERS

An integer is a number with no fractional part.

Integers include:

- Positive numbers
- Zero
- Negative numbers

Examples:

... -3, -2, -1, 0, +1, +2, +3 ...

POSITIVE INTEGERS

Positive integers are numbers greater than zero.

Examples:

+1, +2, +3, +4, +5

The positive sign (+) may sometimes be left out.

Therefore:

+5 = 5

NEGATIVE INTEGERS

Negative integers are numbers less than zero.

Examples:

-1, -2, -3, -4, -5

ZERO

Zero is written as:

0

Zero is neither positive nor negative.

Examples from daily life:

- No money = 0 Frw
- Receiving 50 Frw = +50 Frw
- A shortage of 100 Frw = -100 Frw
- A stone 10 m below the ground = -10 m
- A plane 500 m above the ground = +500 m

EXERCISE 1

Write the correct integer.

1. 5 metres above the ground = _____
2. 7 metres below the ground = _____
3. A gain of 200 Frw = _____
4. A loss of 300 Frw = _____
5. No marks scored = _____
6. A pupil climbs 10 metres up a tree = _____
7. A coin is 2 metres below the surface of a swimming pool = _____

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3. NUMBER LINE

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A number line is a horizontal line on which integers are marked.

Example:

-5 -4 -3 -2 -1 0 +1 +2 +3 +4 +5

RULES:

- Zero is the reference point.
- Positive numbers are found to the RIGHT of zero.
- Negative numbers are found to the LEFT of zero.
- Moving to the right gives greater numbers.
- Moving to the left gives smaller numbers.

Example:

-3 is to the left of 0.

+4 is to the right of 0.

EXERCISE 2

Draw a number line from -10 to +10.

Then answer:

1. Which integer is 4 steps to the right of +4?
2. Which integer is 5 steps to the left of 0?
3. Which integer is 6 steps to the left of +1?
4. Starting from 0, move 9 steps to the left.
Where do you stop?
5. An insect moves from -6 to +6.
How many steps does it move?

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4. OPPOSITE OF AN INTEGER

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The opposite of an integer has the same value but a different sign.

Examples:

Opposite of +1 = -1

Opposite of +5 = -5

Opposite of -2 = +2

Opposite of -6 = +6

Opposite of +7 = -7

The opposite of an integer is also called its ADDITIVE INVERSE.

IMPORTANT:

A number added to its opposite gives zero.

Example:

$$(+5) + (-5) = 0$$

EXERCISE 3

Write the opposite of each integer.

1. +12 = _____

2. +45 = _____

3. -34 = _____

4. -20 = _____

5. -240 = _____

6. -500 = _____

7. $+75 =$ _____

8. $-1,000 =$ _____

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5. COMPARING POSITIVE AND NEGATIVE INTEGERS

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We can compare integers using a number line.

Numbers become greater as we move to the RIGHT.

Numbers become smaller as we move to the LEFT.

Examples:

$$+5 > -2$$

because +5 is to the right of -2.

$$-1 > -6$$

because -1 is to the right of -6.

$$-10 < -9$$

$$-7 < -4$$

$$-2 < 0$$

$$+1 < +6$$

$$0 > -4$$

$$+3 > -10$$

REMEMBER:

Among negative numbers, the number closer to zero is greater.

Example:

$$-2 > -8$$

EXERCISE 4

Put $<$, $>$ or $=$.

1. -5 _____ $+3$

2. -1 _____ -6

3. $+4$ _____ -2

4. -8 _____ -3

5. 0 _____ -7

6. $+6$ _____ $+10$

7. -10 _____ -9

8. $+5$ _____ $+5$

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6. ORDERING INTEGERS

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ASCENDING ORDER means arranging from the smallest to the greatest.

Example:

$+1, -3, -10, -2, -4, 0, +5$

Ascending order:

$-10, -4, -3, -2, 0, +1, +5$

DESCENDING ORDER means arranging from the greatest to the smallest.

Example:

$-3, +6, 0, -8, +2$

Descending order:

+6, +2, 0, -3, -8

EXERCISE 5

1. Arrange from smallest to greatest:

-2, +1, -4, +3

2. Arrange from greatest to smallest:

-5, +4, 0, -2, +8

3. Arrange in ascending order:

-8, -5, +3, +2, 0, +5, +6, +8, +1

4. Arrange in descending order:

-3, -7, 0, +4, -13, +13, +31

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7. DISTANCE BETWEEN INTEGERS
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The magnitude of a number is its distance from zero.

Example:

Magnitude of +3 = 3

Magnitude of -3 = 3

A. INTEGERS ON THE SAME SIDE OF ZERO

If two integers are on the same side of zero, subtract their magnitudes.

Example:

Distance between +2 and +10:

$$10 - 2 = 8 \text{ steps}$$

Distance between -3 and -6:

$$6 - 3 = 3 \text{ steps}$$

Distance between -7 and -2:

$$7 - 2 = 5 \text{ steps}$$

B. INTEGERS ON OPPOSITE SIDES OF ZERO

If the integers are on opposite sides of zero, add their magnitudes.

Example:

Distance between -5 and +7:

$$5 + 7 = 12 \text{ steps}$$

Distance between -10 and +10:

$$10 + 10 = 20 \text{ steps}$$

----- EXERCISE 6 -----

Find the distance between:

1. 0 and +3

2. 0 and -3

3. -2 and +2

4. -1 and +1

5. +3 and -1

6. -2 and +10

7. -10 and +10

8. -10 and -6

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8. SOLVING PROBLEMS USING INTEGERS

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Integers help us solve problems involving:

- Temperature
- Money
- Buildings
- Heights and depths
- Sports
- Movement above or below a reference point

Example 1:

A car is parked on floor -3.

It moves up 4 floors.

$-3 \rightarrow -2 \rightarrow -1 \rightarrow 0 \rightarrow +1$

Therefore, the car reaches floor +1.

Example 2:

The temperature is 18°C .

It drops by 7°C .

$18 - 7 = 11$

New temperature = 11°C

EXERCISE 7

1. A car is on floor -3 and moves up 5 floors.

Which floor does it reach?

2. The temperature is 18°C and drops by 7°C .

What is the new temperature?

3. A diver is at -8 metres and moves up 3 metres.

What is the diver's new position?

4. A person is at +6 and moves down 10 steps.

Where does the person stop?

5. A lift is on floor -2 and moves up 6 floors.

Which floor does it reach?

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UNIT 2 REVISION EXERCISE

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1. Write the correct integer for each statement.

a) 3 metres below the ground.

b) 5 steps forward.

c) 10 steps backward.

2. Write the opposite of:

a) -5,890

b) +100,000

c) -20,000

3. Put $<$, $>$ or $=$.

a) -3 _____ $+2$

b) -8 _____ -4

c) 0 _____ -5

d) $+7$ _____ $+3$

4. Arrange in descending order:

$-3, -7, 0, +4, -13, -31, +13, +31, -301, +310, +301$

5. Arrange in ascending order:

$-8, -5, +3, +2, 0, +5, +6, +8, +1$

6. Find the distance between:

a) $+3$ and -1

b) -2 and $+10$

c) -10 and $+10$

d) -10 and -6

7. Write whether each number is positive, negative or neither.

a) $+13$

b) -22

c) 0

d) -1

e) $+100$

8. What is the opposite of +25?

9. What is the opposite of -50?

10. Explain one way negative numbers are used in everyday life.

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QUICK REVISION

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- Integers include negative numbers, zero and positive numbers.
- Positive integers are greater than zero.
- Negative integers are less than zero.
- Zero is neither positive nor negative.
- Positive numbers are to the right of zero on a number line.
- Negative numbers are to the left of zero.
- Numbers increase as we move to the right.
- Numbers decrease as we move to the left.
- The opposite of +5 is -5.
- The opposite of -5 is +5.
- A number plus its opposite equals zero.
- Magnitude means the distance of a number from zero.
- For two integers on the same side of zero, find their distance by subtracting their magnitudes.
- For integers on opposite sides of zero, find their distance by adding their magnitudes.

END OF UNIT 2

UNIT 3: CLASSIFYING NUMBERS BY THEIR PROPERTIES

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1. NATURAL NUMBERS AND WHOLE NUMBERS

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Natural numbers are numbers used for counting.

Examples:

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, ...

Natural numbers:

- Start from 1.
- Do not include zero.
- Do not include negative numbers.
- Do not include fractions.

WHOLE NUMBERS

Whole numbers are natural numbers together with zero.

Examples:

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, ...

REMEMBER:

Natural numbers:

1, 2, 3, 4, 5, ...

Whole numbers:

0, 1, 2, 3, 4, 5, ...

EXERCISE 1

1. Which of these are natural numbers?

2, -5, 3.5, 8, 0, 14, 7.2, -1, 10

2. Write the first 10 natural numbers.

3. Write the first 10 whole numbers.

4. Is 0 a natural number?

5. Is -4 a natural number?

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2. EVEN AND ODD NUMBERS

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EVEN NUMBERS

Even numbers are numbers exactly divisible by 2.

They leave no remainder when divided by 2.

Examples:

2, 4, 6, 8, 10, 12, 14, 16, 18, ...

ODD NUMBERS

Odd numbers are not exactly divisible by 2.

When divided by 2, they leave a remainder of 1.

Examples:

1, 3, 5, 7, 9, 11, 13, 15, 17, 19, ...

REMEMBER:

Even numbers usually end in:

0, 2, 4, 6 or 8

Odd numbers usually end in:

1, 3, 5, 7 or 9

EXERCISE 2

1. List all odd numbers between 0 and 30.

2. List all even numbers from 0 to 30.

3. Write whether each number is EVEN or ODD.

- a) 18
- b) 25
- c) 46
- d) 71
- e) 100
- f) 93

4. Which of these numbers are even?

13, 24, 35, 42, 58, 67, 90

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3. MULTIPLES OF A NUMBER

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Multiples of a number are obtained by multiplying that number by:

1, 2, 3, 4, 5, ...

Example:

Multiples of 2:

- $1 \times 2 = 2$
- $2 \times 2 = 4$
- $3 \times 2 = 6$
- $4 \times 2 = 8$
- $5 \times 2 = 10$

Therefore:

2, 4, 6, 8, 10, 12, ...

Multiples of 5:

5, 10, 15, 20, 25, 30, 35, ...

When we count in twos, fives or tens, we are using multiples.

EXERCISE 3

Write the first 6 multiples of:

1. 2
2. 3
3. 4
4. 5
5. 6
6. 10

7. List all multiples of 3 between 1 and 30.

8. Which numbers between 1 and 30 are NOT multiples of 3?

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4. FACTORS OF A NUMBER

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Factors are whole numbers that divide another number exactly without leaving a remainder.

Factors can also be numbers multiplied together to make another number.

Example:

Factors of 6 are:

1, 2, 3 and 6

Because:

$$1 \times 6 = 6$$

$$2 \times 3 = 6$$

Example:

Find the factors of 18.

$$1 \times 18 = 18$$

$$2 \times 9 = 18$$

$$3 \times 6 = 18$$

Therefore:

Factors of 18:

1, 2, 3, 6, 9, 18

IMPORTANT:

- 1 is the smallest factor of every natural number.
- The number itself is its greatest factor.

EXERCISE 4

List all factors of:

1. 8

2. 12

3. 24

4. 36

5. 48

6. 64

7. 45

8. 23

9. 96

10. 100

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5. PRIME NUMBERS
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A prime number is a number with exactly TWO factors:

- 1
- The number itself

Examples:

2, 3, 5, 7, 11, 13, 17, 19, 23, ...

Example:

Factors of 7:

1 and 7

Therefore, 7 is a prime number.

IMPORTANT:

1 is NOT a prime number because it has only one factor.

The smallest prime number is:

2

2 is also the only even prime number.

EXERCISE 5

1. List the first 10 prime numbers.
2. What is the smallest prime number?
3. Which of these are prime numbers?

2, 4, 5, 9, 11, 15, 17, 21

4. What is the smallest even prime number?
5. What is the smallest odd prime number?

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6. COMPOSITE NUMBERS

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A composite number is a natural number greater than 1 that has more than two factors.

Examples:

4, 6, 8, 9, 10, 12, 14, 15, 16, ...

Example:

Factors of 6:

1, 2, 3, 6

Since 6 has more than two factors, it is a composite number.

REMEMBER:

Prime number = exactly 2 factors

Composite number = more than 2 factors

EXERCISE 6

1. List the first 10 composite numbers.

2. State whether each number is PRIME or COMPOSITE.

- a) 3
- b) 8
- c) 11
- d) 12
- e) 17
- f) 20

3. Which of these are composite numbers?

5, 6, 7, 9, 13, 15, 18

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7. SQUARE NUMBERS

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A square number is obtained when a number is multiplied by itself.

Examples:

$$1 \times 1 = 1$$

$$2 \times 2 = 4$$

$$3 \times 3 = 9$$

$$4 \times 4 = 16$$

$$5 \times 5 = 25$$

$$6 \times 6 = 36$$

$$7 \times 7 = 49$$

$$8 \times 8 = 64$$

$$9 \times 9 = 81$$

$$10 \times 10 = 100$$

Therefore, square numbers up to 100 are:

1, 4, 9, 16, 25, 36, 49, 64, 81, 100

EXERCISE 7

1. Find:

a) 2×2

b) 4×4

c) 6×6

d) 8×8

e) 10×10

2. List all square numbers up to 100.

3. Which of these are square numbers?

9, 12, 16, 20, 25, 36, 45, 49

=====

8. SQUARE ROOT

=====

Finding the square root is the opposite of finding the square of a number.

The square root of a square number is the number that was multiplied by itself.

Examples:

$$3 \times 3 = 9$$

Therefore:

$$\sqrt{9} = 3$$

$$5 \times 5 = 25$$

Therefore:

$$\sqrt{25} = 5$$

$$6 \times 6 = 36$$

Therefore:

$$\sqrt{36} = 6$$

$$9 \times 9 = 81$$

Therefore:

$$\sqrt{81} = 9$$

$$10 \times 10 = 100$$

Therefore:

$$\sqrt{100} = 10$$

EXERCISE 8

Find:

1. $\sqrt{1}$

2. $\sqrt{4}$

3. $\sqrt{16}$

4. $\sqrt{25}$

5. $\sqrt{36}$

6. $\sqrt{49}$

7. $\sqrt{64}$

8. $\sqrt{81}$

9. $\sqrt{100}$

=====

9. LOWEST COMMON MULTIPLE (LCM)

=====

LCM means:

LOWEST COMMON MULTIPLE

The LCM of two numbers is the smallest multiple that is common to both numbers.

Example:

Find the LCM of 4 and 6.

Multiples of 4:

4, 8, 12, 16, 20, 24, ...

Multiples of 6:

6, 12, 18, 24, 30, ...

Common multiples:

12, 24, ...

The smallest common multiple is 12.

Therefore:

LCM of 4 and 6 = 12

STEPS FOR FINDING LCM

1. List multiples of the first number.
2. List multiples of the second number.
3. Find common multiples.
4. Choose the smallest common multiple.

Example:

Find LCM of 3 and 4.

Multiples of 3:

3, 6, 9, 12, 15, ...

Multiples of 4:

4, 8, 12, 16, ...

LCM = 12

Example:

Find LCM of 5 and 12.

Multiples of 5:

5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, ...

Multiples of 12:

12, 24, 36, 48, 60, ...

LCM = 60

EXERCISE 9

Find:

1. LCM of 2 and 3
2. LCM of 5 and 10
3. LCM of 3 and 7
4. LCM of 8 and 12
5. LCM of 6 and 9
6. LCM of 4 and 5
7. LCM of 7 and 14
8. LCM of 9 and 12
9. LCM of 10 and 15

=====

10. GREATEST COMMON FACTOR (GCF)

=====

GCF means:

GREATEST COMMON FACTOR

The GCF of two numbers is the largest number that can divide both numbers without leaving a remainder.

STEPS FOR FINDING GCF

1. List all factors of the first number.

2. List all factors of the second number.
3. Identify the common factors.
4. Choose the greatest common factor.

Example:

Find the GCF of 12 and 18.

Factors of 12:

1, 2, 3, 4, 6, 12

Factors of 18:

1, 2, 3, 6, 9, 18

Common factors:

1, 2, 3, 6

The greatest is 6.

Therefore:

GCF of 12 and 18 = 6

Example:

Find GCF of 20 and 30.

Factors of 20:

1, 2, 4, 5, 10, 20

Factors of 30:

1, 2, 3, 5, 6, 10, 15, 30

Common factors:

1, 2, 5, 10

Therefore:

GCF of 20 and 30 = 10

EXERCISE 10

Find the GCF of:

1. 9 and 15
2. 16 and 24
3. 14 and 21
4. 25 and 35
5. 36 and 48
6. 18 and 27

=====

UNIT 3 REVISION EXERCISE

=====

1. Write the first five multiples of:

- a) 3
- b) 6
- c) 5
- d) 9
- e) 10

2. List the first 10 prime numbers.

3. Find the LCM of:

- a) 2 and 4
- b) 4 and 5

c) 3 and 6

d) 4 and 10

e) 12 and 14

f) 23 and 46

4. Find all factors of:

a) 12

b) 18

c) 24

5. Find the GCF of:

a) 12 and 18

b) 20 and 30

c) 16 and 24

6. State whether each number is ODD or EVEN.

a) 25

b) 36

c) 47

d) 80

7. State whether each number is PRIME or COMPOSITE.

a) 7

b) 9

c) 13

d) 16

8. Find:

a) 4^2

b) 6^2

c) 9^2

d) 10^2

9. Find:

a) $\sqrt{16}$

b) $\sqrt{36}$

c) $\sqrt{64}$

d) $\sqrt{100}$

10. Mary has 27 oranges.

a) Is 27 a multiple of 3?

b) If 18 more oranges are added, how many oranges are there altogether?

c) If 9 oranges are given away, how many remain?

=====

QUICK REVISION

=====

- Natural numbers are counting numbers starting from 1.
- Whole numbers include 0 and all natural numbers.
- Even numbers are exactly divisible by 2.
- Odd numbers leave a remainder of 1 when divided by 2.
- Multiples are found by multiplying a number by 1, 2, 3, 4, ...

- Factors divide a number exactly without leaving a remainder.
- A prime number has exactly two factors: 1 and itself.
- 1 is not a prime number.
- A composite number has more than two factors.
- A square number is obtained by multiplying a number by itself.
- Square root is the inverse of squaring.
- LCM is the smallest common multiple of two numbers.
- GCF is the greatest factor common to two numbers.

END OF UNIT 3

UNIT 4: FRACTIONS WITH THE SAME DENOMINATOR

=====

1. INTRODUCTION TO FRACTIONS

=====

A fraction shows a part of a whole.

Example:

If a whole object is divided into 6 equal parts, each part is:

$\frac{1}{6}$

This is read as "one-sixth".

A fraction has two main parts:

Numerator
3

4
Denominator

NUMERATOR

The numerator is the top number.

It tells us how many equal parts we are considering.

DENOMINATOR

The denominator is the bottom number.

It tells us the total number of equal parts that make the whole.

Example:

$$\frac{3}{4}$$

Numerator = 3

Denominator = 4

This means that the whole has been divided into 4 equal parts and we are considering 3 parts.

----- EXERCISE 1 -----

For each fraction, identify the numerator and denominator.

1. $\frac{3}{5}$

2. $\frac{4}{7}$

3. $\frac{2}{9}$

4. $\frac{7}{10}$

5. $\frac{11}{12}$

6. What fraction represents one part when a whole is divided into 8 equal parts?

7. An orange is divided into 8 equal pieces. One piece is eaten.
What fraction was eaten?

8. What fraction remains?

=====

2. READING AND WRITING FRACTIONS

=====

To read a fraction:

- Read the numerator first.
- Then read the denominator using its fraction name.

Examples:

$1/2$ = one-half

$1/3$ = one-third

$1/4$ = one-fourth or one-quarter

$2/5$ = two-fifths

$4/5$ = four-fifths

$1/10$ = one-tenth

$7/1000$ = seven-thousandths

Example:

$3/4$

is read as:

Three-fourths

EXERCISE 2

A. Write these fractions in words.

1. $1/2$
2. $2/3$
3. $3/4$
4. $4/5$
5. $7/10$

B. Write these in figures.

1. One-half
2. One-third

3. Three-fourths
4. Two-fifths
5. Seven-eighths

=====

3. SIMPLIFYING FRACTIONS

=====

Simplifying a fraction means writing it in its simplest form without changing its value.

To simplify a fraction:

1. Find a number that divides both the numerator and denominator.
2. It is best to use their GCF.
3. Divide both numerator and denominator by that number.

Example 1:

Simplify:

$$\frac{4}{8}$$

$$\text{GCF of 4 and 8} = 4$$

$$4 \div 4 = 1$$

$$8 \div 4 = 2$$

Therefore:

$$\frac{4}{8} = \frac{1}{2}$$

Example 2:

$$\frac{3}{9}$$

$$\text{GCF of 3 and 9} = 3$$

$$3 \div 3 = 1$$

$$9 \div 3 = 3$$

Therefore:

$$\frac{3}{9} = \frac{1}{3}$$

Example 3:

$$2/10$$

$$2 \div 2 = 1$$

$$10 \div 2 = 5$$

Therefore:

$$2/10 = 1/5$$

A fraction is in its simplest form when only 1 can divide both the numerator and denominator exactly.

EXERCISE 3

Simplify:

1. $10/15$

2. $12/36$

3. $18/30$

4. $20/28$

5. $4/8$

6. $6/12$

7. $8/16$

WORD PROBLEMS

8. You ate 4 slices of pizza out of 8 slices.

Write and simplify the fraction eaten.

9. Three pencils out of six pencils are broken.

Write and simplify the fraction of broken pencils.

10. Two oranges out of ten are rotten.

Write and simplify the fraction of rotten oranges.

=====

4. COMPARING FRACTIONS WITH THE SAME DENOMINATOR

=====

When fractions have the same denominator, compare their numerators.

The fraction with the greater numerator is greater.

Example:

$\frac{10}{17}$ and $\frac{8}{17}$

Since:

$$10 > 8$$

Therefore:

$$\frac{10}{17} > \frac{8}{17}$$

Example:

$\frac{2}{3}$ and $\frac{1}{3}$

$$2 > 1$$

Therefore:

$$\frac{2}{3} > \frac{1}{3}$$

Example:

$\frac{1}{9}$ and $\frac{4}{9}$

$$1 < 4$$

Therefore:

$$\frac{1}{9} < \frac{4}{9}$$

The signs used are:

$>$ means greater than

$<$ means less than

$=$ means equal to

EXERCISE 4

Put $>$, $<$ or $=$.

1. $\frac{1}{4}$ _____ $\frac{3}{4}$

2. $\frac{6}{7}$ _____ $\frac{2}{7}$

3. $\frac{2}{8}$ _____ $\frac{6}{8}$

4. $\frac{7}{18}$ _____ $\frac{6}{18}$

5. $\frac{2}{11}$ _____ $\frac{5}{11}$

6. $\frac{2}{5}$ _____ $\frac{3}{5}$

7. $\frac{5}{6}$ _____ $\frac{1}{6}$

8. $\frac{3}{4}$ _____ $\frac{3}{4}$

=====

5. ADDITION OF FRACTIONS WITH THE SAME DENOMINATOR

=====

To add fractions with the same denominator:

1. Add the numerators.
2. Keep the denominator the same.

Example:

$$\frac{1}{5} + \frac{2}{5}$$

$$= \frac{(1 + 2)}{5}$$

$$= 3/5$$

Another example:

$$8/16 + 3/16$$

$$= 11/16$$

IMPORTANT:

Do NOT add the denominators when they are the same.

EXERCISE 5

Add:

1. $2/4 + 1/4$

2. $3/7 + 3/7$

3. $4/12 + 3/12$

4. $2/9 + 3/9$

5. $4/10 + 3/10$

6. $2/8 + 5/8$

WORD PROBLEM

7. Manzi ate $1/5$ of a cake at breakfast and $3/5$ at lunch.

What fraction of the cake did he eat altogether?

=====

6. SUBTRACTION OF FRACTIONS WITH THE SAME DENOMINATOR

=====

To subtract fractions with the same denominator:

1. Subtract the numerators.
2. Keep the denominator the same.

Example:

$$\begin{aligned} &5/8 - 2/8 \\ &= (5 - 2)/8 \\ &= 3/8 \end{aligned}$$

Example:

$$\begin{aligned} &6/7 - 2/7 \\ &= 4/7 \end{aligned}$$

EXERCISE 6

Subtract:

1. $5/6 - 2/6$
2. $9/10 - 2/10$
3. $3/4 - 2/4$
4. $2/3 - 1/3$
5. $42/40 - 32/40$
6. $11/12 - 10/12$

=====
7. WORD PROBLEMS ON ADDITION AND SUBTRACTION
=====

Steps:

1. Read the problem carefully.

2. Identify the fractions.
3. Check that the denominators are the same.
4. Add or subtract the numerators.
5. Keep the denominator.
6. Check whether the answer makes sense.

Example 1:

Jane ate $\frac{1}{3}$ of an apple in the morning and $\frac{1}{3}$ in the afternoon.

$$\frac{1}{3} + \frac{1}{3}$$

$$= \frac{2}{3}$$

Jane ate $\frac{2}{3}$ of the apple.

Example 2:

Amos had $\frac{11}{13}$ of a cake and ate $\frac{7}{13}$.

$$\frac{11}{13} - \frac{7}{13}$$

$$= \frac{4}{13}$$

$\frac{4}{13}$ of the cake remained.

EXERCISE 7

1. A boy drinks $\frac{1}{2}$ of a glass of milk.

What fraction remains?

2. Adults occupy $\frac{2}{3}$ of the seats on a bus.

What fraction is occupied by children?

3. A piece of land is divided into 8 equal parts.

Marc takes $\frac{2}{8}$ and Jean takes $\frac{4}{8}$.

What fraction remains for Bella?

4. A mother gives $\frac{1}{5}$ of her sugar cane to her daughter and $\frac{4}{5}$ to her son.

What fraction did she give altogether?

=====

8. MULTIPLICATION OF A FRACTION BY A WHOLE NUMBER

=====

To multiply a fraction by a whole number:

- Multiply the numerator by the whole number.
- Keep the denominator the same.
- Simplify if necessary.

Example:

$$3 \times \frac{1}{4}$$

$$= \frac{3}{4}$$

A whole number can also be written over 1.

Example:

$$3 = \frac{3}{1}$$

Therefore:

$$\frac{3}{1} \times \frac{1}{4}$$

$$= \frac{3}{4}$$

EXERCISE 8

Calculate:

1. $\frac{2}{3} \times 5$

2. $\frac{3}{5} \times 10$

3. $\frac{8}{10} \times 5$

4. $\frac{1}{2} \times 2$

5. $\frac{1}{2} \times 5$

6. $\frac{9}{10} \times 12$

=====

9. MULTIPLICATION OF FRACTIONS BY FRACTIONS

=====

To multiply two fractions:

1. Multiply the numerators.
2. Multiply the denominators.
3. Simplify the answer if possible.

Example:

$$\frac{2}{3} \times \frac{2}{5}$$

Numerators:

$$2 \times 2 = 4$$

Denominators:

$$3 \times 5 = 15$$

Therefore:

$$\frac{2}{3} \times \frac{2}{5} = \frac{4}{15}$$

EXERCISE 9

Multiply and simplify where necessary.

1. $\frac{1}{3} \times \frac{1}{4}$

2. $\frac{2}{3} \times \frac{1}{3}$

3. $\frac{4}{5} \times \frac{1}{7}$

4. $\frac{2}{7} \times \frac{1}{5}$

5. $\frac{3}{8} \times \frac{1}{2}$

6. $\frac{3}{4} \times \frac{1}{6}$

WORD PROBLEMS

7. A lesson lasts 40 minutes.

If $\frac{1}{8}$ of the lesson is used for revision, how many minutes are used for revision?

8. A ribbon is $\frac{2}{3}$ of a metre long.

Sarah cuts off $\frac{1}{4}$ of the ribbon.

What fraction of a metre does she cut off?

9. John read $\frac{2}{5}$ of a book on Monday.

On Tuesday he read $\frac{1}{2}$ of what he read on Monday.

What fraction of the whole book did he read on Tuesday?

=====

10. DIVISION OF A FRACTION BY A WHOLE NUMBER

=====

Dividing a fraction by a whole number means splitting the fraction into equal parts.

The book gives these steps:

1. Keep the numerator.
2. Multiply the denominator by the whole number.

3. Write the new fraction.
4. Simplify if necessary.

Example:

$$1/2 \div 3$$

$$= 1/(2 \times 3)$$

$$= 1/6$$

Therefore:

$$1/2 \div 3 = 1/6$$

EXERCISE 10

Calculate:

1. $1/6 \div 7$

2. $2/3 \div 7$

3. $1/6 \div 9$

4. $4/5 \div 2$

5. $1/8 \div 5$

6. $2/3 \div 5$

WORD PROBLEMS

7. Half of a pawpaw is shared equally between 2 people.

What fraction of the whole pawpaw does each person receive?

8. You have $2/3$ of a carton of juice and share it equally between 2 friends.

What fraction of the whole carton does each friend receive?

=====

UNIT 4 REVISION EXERCISE

=====

1. Name the numerator and denominator of:

a) $\frac{3}{8}$

b) $\frac{5}{9}$

2. Write in words:

a) $\frac{1}{2}$

b) $\frac{3}{5}$

c) $\frac{7}{10}$

3. Simplify:

a) $\frac{4}{8}$

b) $\frac{6}{12}$

c) $\frac{10}{15}$

d) $\frac{12}{36}$

4. Put $>$, $<$ or $=$.

a) $\frac{3}{8}$ _____ $\frac{1}{8}$

b) $\frac{4}{9}$ _____ $\frac{5}{9}$

c) $\frac{7}{12}$ _____ $\frac{6}{12}$

d) $\frac{5}{10}$ _____ $\frac{5}{10}$

5. Add:

a) $\frac{3}{10} + \frac{4}{10}$

b) $\frac{1}{5} + \frac{3}{5}$

c) $\frac{4}{9} + \frac{1}{9}$

d) $\frac{3}{6} + \frac{2}{6}$

6. Subtract:

a) $\frac{4}{5} - \frac{1}{5}$

b) $\frac{4}{7} - \frac{3}{7}$

c) $\frac{7}{15} - \frac{5}{15}$

d) $\frac{125}{125} - \frac{120}{125}$

7. Multiply:

a) $\frac{4}{5} \times 3$

b) $\frac{1}{8} \times 3$

c) $\frac{1}{2} \times 9$

d) $\frac{1}{4} \times 8$

8. Multiply:

a) $\frac{8}{9} \times \frac{3}{4}$

b) $\frac{9}{11} \times \frac{1}{3}$

c) $\frac{2}{5} \times \frac{4}{9}$

9. Find:

$\frac{4}{7}$ of 14

10. A class has 30 pupils.

Two-thirds of them are girls.

How many girls are in the class?

11. A stadium has 20,000 spectators.

Half of them are wearing red jerseys.

How many spectators are wearing red jerseys?

12. A mother shares one cake equally among 5 children.

What fraction does each child receive?

13. A father earns 25,000 Frw per month.

He spends:

$\frac{1}{5}$ on food

$\frac{3}{5}$ on school fees

and saves the rest.

Find:

a) Amount spent on food.

b) Amount spent on school fees.

c) Amount saved.

=====

QUICK REVISION

=====

- A fraction represents part of a whole.
- Numerator = top number.
- Denominator = bottom number.
- The denominator tells the total number of equal parts.

- The numerator tells how many parts are being considered.
- Simplifying means writing an equivalent fraction in its simplest form.
- To compare fractions with the same denominator, compare their numerators.
- To add fractions with the same denominator, add the numerators and keep the denominator.
- To subtract fractions with the same denominator, subtract the numerators and keep the denominator.
- To multiply fractions, multiply numerators together and denominators together.
- To multiply a fraction by a whole number, multiply the numerator by the whole number.
- To divide a fraction by a whole number, multiply the denominator by the whole number.
- Always simplify the final answer when possible.

END OF UNIT 4

UNIT 5: DECIMAL NUMBERS

1. MEANING OF DECIMAL NUMBERS

A decimal number has a whole-number part and a decimal part separated by a decimal point.

Example: 28.53

- Whole-number part = 28
- Decimal part = 53

Decimals are used in shopping, measuring, cooking and other daily calculations.

Examples:

1.25 m, 2.75 m, 4.50 Frw, 0.75 L.

2. DECIMALS ON A NUMBER LINE

Between 0 and 1, we can have decimals such as:

0.1, 0.2, 0.3, 0.4, 0.5, ... 0.9

Examples:

$$0.1 = \frac{1}{10}$$

$$0.5 = \frac{5}{10} = \frac{1}{2}$$

$$0.9 = \frac{9}{10}$$

Hundredths include:

$$0.01 = 1/100$$

$$0.02 = 2/100$$

$$0.05 = 5/100$$

$$0.09 = 9/100$$

Zeros at the end of a decimal do not change its value:

$$1.2300 = 1.23$$

$$4.050 = 4.05$$

$$5.00 = 5$$

EXERCISE 1

1. Write as decimals: $3/10$, $7/10$, $4/100$, $9/100$.

2. Which is between 0 and 1: 1.5, 0.6, 2.3?

3. Simplify: 2.500, 4.060, 8.00.

3. PLACE VALUE OF DECIMALS

Example: 78.46

$$7 = \text{tens} = 70$$

$$8 = \text{ones} = 8$$

$$4 = \text{tenths} = 0.4$$

$$6 = \text{hundredths} = 0.06$$

Place-value order:

Tens | Ones | . | Tenths | Hundredths

78.46 is read:

“Seventy-eight point four six.”

We read each digit after the decimal point separately.

EXERCISE 2

Give the place value of the indicated decimal digits:

1. 8.25

2. 17.36

3. 0.47

4. 23.56

5. 87.09

4. COMPARING DECIMAL NUMBERS

Use:

> greater than

< less than

= equal to

Steps:

1. Compare whole-number parts.

2. If equal, compare tenths.
3. If still equal, compare hundredths.

Example:

9.25 and 9.70

Whole parts are equal.

2 tenths < 7 tenths

Therefore: $9.25 < 9.70$

Example:

$452.78 > 452.21$ because 7 tenths > 2 tenths.

You may add zeros to compare:

$2.8 = 2.80$

EXERCISE 3

Insert >, < or =:

1. $1.02 \underline{\hspace{1cm}} 1.12$

2. $1.66 \underline{\hspace{1cm}} 1.70$

3. $3.10 \underline{\hspace{1cm}} 2.49$

4. $2.52 \underline{\hspace{1cm}} 2.80$

5. $87.12 \underline{\hspace{1cm}} 78.12$

6. $4.40 \underline{\hspace{1cm}} 0.44$

7. $9.01 \underline{\hspace{1cm}} 9.01$

Arrange from smallest to largest:

8. 3.5, 3.79, 3.42, 3.57, 3.70, 3.62

5. ADDITION OF DECIMALS

When adding decimals:

- Put decimal points directly under each other.
- Add zeros where necessary.
- Add as you do with whole numbers.

Example:

```
  3.21
+ 4.50
-----
  7.71
```

A whole number can be written with decimal zeros:

$528 = 528.00$

EXERCISE 4

Add:

1. $206.1 + 223.9$

2. $404.1 + 247.4$

3. $665.2 + 567$
4. $435.7 + 65.1$
5. $299.72 + 0.08$
6. $84.0 + 79.3$

6. SUBTRACTION OF DECIMALS

Line up decimal points before subtracting.

Example:

$$8.97 - 2.82 = 6.15$$

Example:

$$3.80 - 1.26 = 2.54$$

EXERCISE 5

Subtract:

1. $18.6 - 8.6$
2. $24.6 - 20.4$
3. $80.02 - 0.02$
4. $19.82 - 5.28$
5. $9.1 - 6.1$
6. $53.3 - 23.2$
7. $689.6 - 609.8$

7. WORD PROBLEMS WITH DECIMALS

Steps:

1. Read the question.
2. Identify the given information.
3. Decide whether to add or subtract.
4. Calculate.
5. Write the answer with the correct unit.

Example:

Alice had 4 m of cloth and used 2.3 m.

Remaining cloth:

$$4.0 - 2.3 = 1.7 \text{ m}$$

EXERCISE 6

1. Mahoro weighs 94.2 kg and Munezero weighs 87.5 kg. Find the difference.
2. A boy walks 2.5 km to school and 1.5 km to his friend's home. Find the total distance.
3. A tailor has 2.75 m of fabric, uses 1.40 m and later buys 3.25 m. How much fabric does she have now?
4. Four rivers measure 2.5 km, 4 km, 10.45 km and 0.95 km.
 - a) Find their total length.
 - b) Find the difference between the longest and shortest.

UNIT 5 REVISION EXERCISE

1. Write the whole-number and decimal parts of 46.27.
2. What is the place value of 5 in 12.56?
3. Write 0.7 as a fraction.
4. Insert $>$, $<$ or $=$:
 - a) 5.65 ____ 5.56
 - b) 6.78 ____ 6.87
 - c) 4.28 ____ 4.82
 - d) 42.7 ____ 41.99
5. Calculate:
 - a) $3.21 + 4.50$
 - b) $206.1 + 223.9$
 - c) $8.97 - 2.82$
 - d) $19.82 - 5.28$
6. Three friends share 1.5 litres of juice equally. How much does each receive?
7. A tap wastes 0.25 litres each hour. How much water is wasted in 24 hours?

QUICK REVISION

- Decimal point separates whole and decimal parts.
- First digit after the decimal = tenths.
- Second digit after the decimal = hundredths.
- $0.5 = 5/10$.
- Ending zeros do not change decimal value.
- Compare decimals from left to right.
- Always line up decimal points when adding or subtracting.
- Add zeros where necessary, e.g. $4.5 = 4.50$.
- Always include the correct unit in word-problem answers.

END OF UNIT 5

UNIT 6: LENGTH MEASUREMENTS

1. MEANING OF LENGTH

Length tells us how long, wide or tall something is.

Examples:

- Length of a pencil
- Height of a person
- Length of a road
- Width of a classroom

Common measuring tools:

- Ruler – short lengths.
- Measuring tape – longer lengths, cloth, height, etc.
- Yard stick – small lengths.
- Decameter – useful for longer distances such as roads or plots.

EXERCISE 1

1. What is length?
2. Name 3 instruments used to measure length.
3. Which tool would you use to measure:
a) a pencil? b) cloth? c) a field?

2. NON-STANDARD UNITS OF LENGTH

These are measurements that are not fixed because they may differ from person to person.

Examples:

- Pace/footstep
- Handspan
- Foot span
- Cubit
- Arm span
- Palm

Example: Jean and Alice may get different answers when measuring a desk with handspans because their hands are different sizes.

EXERCISE 2

1. Name four non-standard units.
2. Which body part is used for a cubit?
3. Why can two people get different measurements using handspans?

3. STANDARD UNITS OF LENGTH

The standard unit of length is the metre (m).

Other metric units:

km = kilometre
hm = hectometre
dam = decametre
m = metre
dm = decimetre
cm = centimetre
mm = millimetre

Important conversions:

1 km = 1,000 m
1 hm = 100 m
1 dam = 10 m
1 m = 10 dm = 100 cm = 1,000 mm
1 dm = 10 cm
1 cm = 10 mm

Conversion order:

km → hm → dam → m → dm → cm → mm

Moving one step RIGHT: $\times 10$

Moving one step LEFT: $\div 10$

Examples:

2 km = 2,000 m

3 dam = 30 m

10 cm = 100 mm

2,000 m = 2 km

30,000 mm = 30 m

EXERCISE 3

Convert:

1. 40 km = _____ m

2. 350 cm = _____ mm

3. 3 hm = _____ m

4. 1 m = _____ mm

5. 2,900 mm = _____ cm

6. 2,000 m = _____ km

7. 46 cm = _____ mm

8. 50 mm = _____ cm

4. ESTIMATING AND MEASURING LENGTH

Estimation means making a sensible guess before measuring.

Example:

Estimated pencil length = 15 cm

Actual measured length = 14.7 cm

How to use a metre ruler:

1. Place the object properly.
2. Start from the 0 mark.
3. Read where the object ends.
4. Record the answer with the correct unit.

A metre ruler is 100 cm long.

EXERCISE 4

Estimate and then measure:

1. Length of your pencil.
2. Width of your book.
3. Length of your desk.
4. Height of your friend.

5. ADDITION AND SUBTRACTION OF LENGTHS

Before adding or subtracting different units, convert them to the SAME unit.

Example:

$$\begin{aligned}
 &3 \text{ km} + 200 \text{ m} \\
 &= 3,000 \text{ m} + 200 \text{ m} \\
 &= 3,200 \text{ m}
 \end{aligned}$$

Example:

$$226 \text{ cm} - 105 \text{ cm} = 121 \text{ cm}$$

Example:

$$\begin{aligned}
 &3 \text{ m } 15 \text{ cm} - 1 \text{ m } 35 \text{ cm} \\
 &= 315 \text{ cm} - 135 \text{ cm} \\
 &= 180 \text{ cm}
 \end{aligned}$$

EXERCISE 5

Calculate:

1. $30 \text{ km} + 4,000 \text{ m} = \underline{\hspace{2cm}} \text{ m}$
2. $65 \text{ hm} - 25 \text{ dam} = \underline{\hspace{2cm}} \text{ m}$
3. $30 \text{ km} - 4,000 \text{ m} = \underline{\hspace{2cm}} \text{ hm}$
4. $500 \text{ m} + 2,800 \text{ dm} = \underline{\hspace{2cm}} \text{ cm}$
5. $250 \text{ cm} - 125 \text{ cm} = \underline{\hspace{2cm}} \text{ cm}$

6. PERIMETER

Perimeter is the total distance around a closed shape.

To find perimeter:

ADD the lengths of all sides.

Example:

A shape has sides 6 cm, 2 cm, 6 cm and 2 cm.

$$\begin{aligned}
 \text{Perimeter} &= 6 + 2 + 6 + 2 \\
 &= 16 \text{ cm}
 \end{aligned}$$

For a rectangle:

$$\text{Perimeter} = 2 \times (\text{Length} + \text{Width})$$

Example:

$$\text{Length} = 12 \text{ cm}$$

$$\text{Width} = 7 \text{ cm}$$

$$\begin{aligned}
 P &= 2 \times (12 + 7) \\
 &= 2 \times 19 \\
 &= 38 \text{ cm}
 \end{aligned}$$

EXERCISE 6

1. Find the perimeter of a rectangle:

$$\text{Length} = 8 \text{ m, Width} = 5 \text{ m.}$$

2. Find the perimeter:

Sides = 5 cm, 7 cm, 5 cm and 7 cm.

3. A garden is 10 m long and 6 m wide. How much fencing is needed around it?

7. REAL-LIFE PROBLEMS

Length is used when measuring roads, land, buildings, cloth, fences and many other things.

Example:

11 bundles of wire are each 125 cm long.

$$\begin{aligned}\text{Total length} &= 11 \times 125 \text{ cm} \\ &= 1,375 \text{ cm}\end{aligned}$$

UNIT 6 REVISION EXERCISE

1. Complete:

- a) 1 km = _____ m
- b) 1 m = _____ cm
- c) 1 cm = _____ mm
- d) 1 hm = _____ m

2. Convert:

- a) 5 km = _____ m
- b) 25 m = _____ cm
- c) 3 km = _____ dam
- d) 5,000 m = _____ km

3. Calculate:

- a) 1,100 mm + 1,100 cm = _____ cm
- b) 3 km + 200 m = _____ m
- c) 226 cm - 105 cm = _____ cm

4. A rectangular garden is 15 m long and 8 m wide.

Find its perimeter.

5. Three pieces of wood measure 85 cm, 120 cm and 95 cm.

Find their total length.

6. Six metal poles are each 150 cm long.

Find their total length.

7. Five garden signs are each 30 cm wide.

Find their total width.

QUICK REVISION

- Length tells how long, wide or tall something is.

- Metre (m) is the standard unit of length.
- Metric order: km, hm, dam, m, dm, cm, mm.
- 1 km = 1,000 m.
- 1 m = 100 cm = 1,000 mm.
- 1 cm = 10 mm.
- Estimate first, then measure accurately.
- Start measuring from the 0 mark.
- Convert to the same unit before adding or subtracting.
- Perimeter = total distance around a shape.
- Rectangle perimeter = $2 \times (\text{Length} + \text{Width})$.

END OF UNIT 6

UNIT 7: CAPACITY MEASUREMENTS

1. MEANING OF CAPACITY

Capacity tells us how much liquid a container can hold.

Common units:

- ml = millilitre
- cl = centilitre
- dl = decilitre
- l = litre
- dal = decalitre
- hl = hectolitre
- kl = kilolitre

Examples:

- Medicine spoon $\approx$ 5 ml
- Cup $\approx$ 50 cl
- Jug $\approx$ 3 l
- Bucket $\approx$ 15 l

Small amounts are usually measured in ml; larger amounts in litres.

:contentReference[oaicite:0]{index=0}

EXERCISE 1

1. What is capacity?
2. Choose a suitable unit: medicine spoon, bucket, juice bottle.
3. Which holds more: 500 ml or 2 l?

2. ESTIMATING AND MEASURING CAPACITY

We can estimate how much liquid a container can hold, then measure it.

Example:

A 5-litre jerrycan can be filled by five 1-litre containers.

EXERCISE 2

Estimate the capacity of:

1. Cup
2. Jug
3. Bucket
4. Jerrycan
5. Water bottle

3. CONVERSION OF CAPACITY

Conversion order:

kl $\rightarrow$ hl $\rightarrow$ dal $\rightarrow$ l $\rightarrow$ dl $\rightarrow$ cl $\rightarrow$ ml

Important:

1 l = 1,000 ml

To change:

- litres $\rightarrow$ millilitres: $\times 1,000$
- millilitres $\rightarrow$ litres: $\div 1,000$
- One step to the right in the table: $\times 10$
- One step to the left: $\div 10$

Example:

5 kl = 500 dal

Example:

60,000 ml = 60 l :contentReference[oaicite:1]{index=1}
:contentReference[oaicite:2]{index=2}

EXERCISE 3

Convert:

1. 50 kl = _____ dal
2. 100 dl = _____ ml
3. 35 hl = _____ dal
4. 7,800 cl = _____ l
5. 60,000 ml = _____ l
6. 4 l = _____ ml
7. 3,500 ml = _____ l

4. ADDITION AND SUBTRACTION OF CAPACITY

Before adding or subtracting different units, convert them to the SAME unit.

Example:

900 dal + 400 l
= 9,000 l + 400 l
= 9,400 l

Example:

$$24 \text{ l} - 2,400 \text{ cl}$$

$$24 \text{ l} = 2,400 \text{ cl}$$

Therefore:

$$2,400 \text{ cl} - 2,400 \text{ cl} = 0 \text{ cl}$$

EXERCISE 4

Calculate:

1. $2 \text{ l} + 7 \text{ l}$

2. $9,340 \text{ l} + 7,280 \text{ l}$

3. $9,730 \text{ cl} - 64 \text{ l}$

4. $10 \text{ l } 7 \text{ dl} - 4 \text{ l } 3 \text{ dl}$

5. $3.45 \text{ l} + 6.65 \text{ l}$

6. $10.5 \text{ l} - 2.75 \text{ l}$

5. COMPARING CAPACITIES

To compare capacities, first convert them to the same unit.

Example:

$$1 \text{ l} = 1,000 \text{ ml}$$

Therefore:

$$1 \text{ l} > 750 \text{ ml}$$

Difference:

$$1,000 \text{ ml} - 750 \text{ ml} = 250 \text{ ml}$$

EXERCISE 5

1. Which is greater: 2 l or 1,500 ml?

2. By how much is 750 ml less than 1 l?

3. Which is greater: 3 l or 2,800 ml?

6. REAL-LIFE PROBLEMS

Example:

A jug holds 2 l. You need 10 l of water.

$$10 \div 2 = 5$$

The jug must be filled 5 times.

Example:

11 cans of milk are sold each week. Each holds 20 l.

In 4 weeks:

$$11 \times 4 = 44 \text{ cans}$$

$$44 \times 20 = 880 \text{ l}$$

Therefore, 880 l are sold in one month. :contentReference[oaicite:3]{index=3}

EXERCISE 6

1. A tank has 12,000 l. Pupils use 40 hl. How much remains?
2. One cup of milk costs 150 Frw. Two cups make 1 litre. Find the cost of 4 litres.
3. A cow produces 15 l in the morning and 10 l in the evening. How much does it produce in 7 days?
4. Fuel costs 1,630 Frw per litre. Find the cost of 50 litres.

UNIT 7 REVISION EXERCISE

1. Complete:

- a) 1 l = _____ ml
- b) 5 l = _____ ml
- c) 2,000 ml = _____ l

2. Convert:

- a) 50 kl to dal
- b) 35 hl to dal
- c) 7,800 cl to l
- d) 60,000 ml to l

3. Add:

- a) 30,500 ml + 50,700 ml
- b) 3.45 l + 6.65 l

4. Subtract:

- a) 5 l 400 ml - 3 l 250 ml
- b) 10.5 l - 2.75 l

5. A water tank has 12,000 l and 40 hl are used. Find the amount remaining.

6. A farmer sells 200 l of milk every day. How much milk is sold in 90 days?

QUICK REVISION

- Capacity = amount of liquid a container can hold.
- Units: kl, hl, dal, l, dl, cl, ml.
- 1 l = 1,000 ml.
- Convert to the same unit before adding, subtracting or comparing.
- Larger containers are usually measured in litres.
- Small liquid amounts are usually measured in millilitres.
- Capacity is useful when measuring water, milk, fuel, juice and other liquids.

END OF UNIT 7

UNIT 8: MASS MEASUREMENTS

1. MEANING OF MASS

Mass tells us how heavy or light an object is.

Examples:

- Light objects such as a pencil, egg or tomato are usually measured in grams (g).
- Heavier objects such as a chair, watermelon or schoolbag are usually measured in kilograms (kg).
- Very heavy objects such as cars and machines can be measured in tonnes (t).

Instruments used to measure mass include:

- Top pan balance
- Beam balance
- Triple beam balance
- Electronic balance

The standard unit of mass is the kilogram (kg).

EXERCISE 1

1. What is mass?
2. Name two instruments used to measure mass.
3. Choose g or kg for:
a) Egg b) Bicycle c) Pencil d) Watermelon

2. ESTIMATING AND MEASURING MASS

We can first estimate whether an object is heavy or light, then use a balance to find its exact mass.

Steps:

1. Estimate the mass.
2. Place the object on a balance/scale.
3. Read the measurement.
4. Record it with the correct unit.

EXERCISE 2

Choose a suitable unit (g, kg or t):

1. Book
2. Bus
3. Orange
4. Cow
5. Bag of rice
6. Truck

3. UNITS OF MASS

The main units are:

t = tonne
kg = kilogram
hg = hectogram
dag = decagram

g = gram
dg = decigram
cg = centigram
mg = milligram

Important conversions:

1 t = 1,000 kg
1 kg = 1,000 g
1 hg = 100 g
1 dag = 10 g
1 dg = 0.1 g
1 cg = 0.01 g
1 mg = 0.001 g

4. CONVERSION OF MASS

To convert:

- kg → g: multiply by 1,000
- g → kg: divide by 1,000

Examples:

2 kg = 2,000 g
2 kg = 20 hg
40 g = 40,000 mg
50 dg = 500 cg

EXERCISE 3

Convert:

1. 24 kg = _____ g
2. 8 kg = _____ g
3. 1.5 kg = _____ g
4. 15,000 g = _____ kg
5. 2,400 g = _____ kg
6. 500 g = _____ kg
7. 3 t = _____ kg

5. ADDITION AND SUBTRACTION OF MASS

Before adding or subtracting masses, convert them to the SAME unit.

Example:

4 kg + 3 hg
3 hg = 0.3 kg
4 + 0.3 = 4.3 kg

Example:

10 t - 9,600 kg
10 t = 10,000 kg
10,000 - 9,600 = 400 kg

EXERCISE 4

1. $3.25 \text{ kg} + 1.75 \text{ kg} = \underline{\hspace{2cm}} \text{ g}$
2. $25.65 \text{ kg} - 5.25 \text{ kg} = \underline{\hspace{2cm}} \text{ g}$
3. $4 \text{ kg} + 500 \text{ g} = \underline{\hspace{2cm}} \text{ g}$
4. $8 \text{ kg} - 2,000 \text{ g} = \underline{\hspace{2cm}} \text{ kg}$

6. MULTIPLICATION AND DIVISION OF MASS

Multiplication is useful when several objects have the same mass.

Example:

20 tins each weigh 2 kg 500 g.

$$2 \text{ kg } 500 \text{ g} = 2.5 \text{ kg}$$

$$\text{Total} = 2.5 \times 20 = 50 \text{ kg}$$

Division is used when a total mass is shared equally.

Example:

24 kg of sugar is shared equally into 8 bags.

$$24 \div 8 = 3 \text{ kg}$$

$$3 \text{ kg} = 3,000 \text{ g per bag.}$$

EXERCISE 5

1. One toy weighs 600 g. Find the mass of 10 toys.
2. 20 bags each weigh 5 kg. Find their total mass.
3. Share 30 kg equally among 5 people.
4. 12 kg is packed equally into 4 bags. Find the mass of each bag.

7. AVERAGE MASS

Average mass =

Total mass of all objects $\div$ Number of objects

Example:

Masses = 50 g, 100 g, 150 g and 200 g

$$\text{Total} = 500 \text{ g}$$

$$\text{Number} = 4$$

$$\text{Average} = 500 \div 4 = 125 \text{ g}$$

EXERCISE 6

1. Find the average of 10 kg, 20 kg and 30 kg.
2. Four objects weigh 100 g, 200 g, 300 g and 400 g. Find their average mass.

8. REAL-LIFE PROBLEMS

Mass measurements are important in shopping, transport, cooking and farming.

Example:

A bus can carry a maximum passenger mass of 1,140 kg.
20 passengers each weigh 60 kg.

Total mass = $20 \times 60 = 1,200$ kg

Since $1,200 \text{ kg} > 1,140 \text{ kg}$, the bus is overloaded.

EXERCISE 7

1. A truck carries 200 bags of cement weighing 10 t altogether. Find the mass of one bag.
2. One cup weighs 2 kg 120 g and another weighs 2 kg 865 g. Find the difference.
3. You buy 4 kg of mangoes and eat 2,000 g. How many grams remain?

UNIT 8 REVISION EXERCISE

1. Convert to grams:
a) 285 kg b) 19 kg c) 2.5 kg d) 196 kg
2. Convert to kilograms:
a) 2,126 g b) 9,065 g c) 850 g
3. A car weighs 3 t. Express its mass in kg.
4. A toy car weighs 600 g. Find the total mass of 60 toy cars.
5. 24 kg of sugar is packed equally into 8 bags. How many grams are in each bag?
6. Potatoes weigh 3 kg 420 g. Onions weigh 1,050 g less.
Find the total mass of the potatoes and onions.

QUICK REVISION

- Mass tells how heavy or light an object is.
- Kilogram (kg) is the standard unit of mass.
- Light objects are commonly measured in grams.
- Very heavy objects may be measured in tonnes.
- $1 \text{ t} = 1,000 \text{ kg}$.
- $1 \text{ kg} = 1,000 \text{ g}$.
- Convert to the same unit before adding or subtracting.
- Total mass = mass of one object $\times$ number of objects.
- Average mass = total mass $\div$ number of objects.
- Scales and balances are used to measure mass.

END OF UNIT 8

UNIT 9: AREA AND LAND MEASUREMENTS

1. MEANING OF AREA

Area is the amount of space inside a shape or piece of land.

A unit square is a small square used to measure area.

Example: $1\text{ cm} \times 1\text{ cm} = 1\text{ cm}^2$.

Area is useful when measuring floors, gardens, farms, playgrounds and land.

For a rectangle:

Area = Length $\times$ Width

Example:

Garden: $5\text{ m} \times 4\text{ m}$

Area = 20 m^2 .

EXERCISE 1

1. What is area?
2. Find the area of:
a) $6\text{ m} \times 4\text{ m}$ b) $8\text{ cm} \times 3\text{ cm}$
3. A floor is 10 m long and 5 m wide. Find its area.

2. UNITS OF AREA

The standard unit of area is the square metre (m^2).

Other units:

km^2 , hm^2 , dam^2 , m^2 , dm^2 , cm^2 , mm^2

Order:

$\text{km}^2 \rightarrow \text{hm}^2 \rightarrow \text{dam}^2 \rightarrow \text{m}^2 \rightarrow \text{dm}^2 \rightarrow \text{cm}^2 \rightarrow \text{mm}^2$

For each step:

- Moving right: $\times 100$
- Moving left: $\div 100$

Important:

$1\text{ hm}^2 = 10,000\text{ m}^2$

$1\text{ dam}^2 = 100\text{ m}^2$

$1\text{ m}^2 = 10,000\text{ cm}^2$

$1\text{ cm}^2 = 100\text{ mm}^2$. :contentReference[oaicite:0]{index=0}

EXERCISE 2

Convert:

1. $1\text{ km}^2 = \underline{\hspace{2cm}}\text{ dam}^2$
2. $10\text{ m}^2 = \underline{\hspace{2cm}}\text{ cm}^2$
3. $25\text{ dam}^2 = \underline{\hspace{2cm}}\text{ m}^2$
4. $3\text{ m}^2 = \underline{\hspace{2cm}}\text{ cm}^2$
5. $5\text{ km}^2 = \underline{\hspace{2cm}}\text{ hm}^2$

3. LAND MEASUREMENTS

Large pieces of land can be measured using:

- Hectare (ha)
- Are (a)
- Centiare (ca)

Important conversions:

$$1 \text{ ha} = 100 \text{ a}$$

$$1 \text{ a} = 100 \text{ ca}$$

$$1 \text{ ha} = 10,000 \text{ ca}$$

$$1 \text{ a} = 100 \text{ m}^2$$

$$1 \text{ ca} = 1 \text{ m}^2$$

Also:

$$1 \text{ hm}^2 = 1 \text{ ha}$$

$$1 \text{ dam}^2 = 1 \text{ a}$$

$$1 \text{ m}^2 = 1 \text{ ca. :contentReference[oaicite:1]{index=1} :contentReference[oaicite:2]{index=2}}$$

Example:

$$5,000 \text{ m}^2 = 0.5 \text{ ha}$$

EXERCISE 3

Convert:

$$1. 1 \text{ a} = \underline{\hspace{2cm}} \text{ ca}$$

$$2. 10 \text{ a} = \underline{\hspace{2cm}} \text{ ha}$$

$$3. 25 \text{ m}^2 = \underline{\hspace{2cm}} \text{ ca}$$

$$4. 25 \text{ ca} = \underline{\hspace{2cm}} \text{ a}$$

$$5. 1 \text{ ha} = \underline{\hspace{2cm}} \text{ m}^2$$

$$6. 10 \text{ a} = \underline{\hspace{2cm}} \text{ m}^2$$

$$7. 2.5 \text{ ha} = \underline{\hspace{2cm}} \text{ m}^2$$

4. AREA OF A SQUARE

A square has four equal sides.

$$\text{Area} = \text{Side} \times \text{Side}$$

Example:

$$\text{Side} = 6 \text{ m}$$

$$\text{Area} = 6 \times 6 = 36 \text{ m}^2$$

If area is known:

$$\text{Side} = \sqrt{\text{Area}}$$

Example:

$$\text{Area} = 49 \text{ m}^2$$

$$\text{Side} = \sqrt{49} = 7 \text{ m}$$

5. AREA OF A RECTANGLE

Area = Length $\times$ Width

Example:

Length = 100 m

Width = 40 m

$$\text{Area} = 100 \times 40 = 4,000 \text{ m}^2$$

If area and width are known:

$$\text{Length} = \text{Area} \div \text{Width}$$

If area and length are known:

$$\text{Width} = \text{Area} \div \text{Length}$$

EXERCISE 4

1. Square side = 8 m. Find its area.
2. Rectangle: 50 m $\times$ 20 m. Find its area.
3. A house floor is 10 m $\times$ 8 m. Find its area.
4. A classroom has area 40 m² and width 5 m. Find its length.
5. A rectangular field has area 1,000 m² and width 20 m. Find its length.

6. ADDITION AND SUBTRACTION OF AREAS

Before adding or subtracting different area units, convert them to the same unit.

Example:

$$57.03 \text{ a} + 10.07 \text{ a} = 67.10 \text{ a}$$

Example:

$$16.67 \text{ m}^2 - 9.23 \text{ m}^2 = 7.44 \text{ m}^2$$

Addition combines areas.

Subtraction finds the area remaining or the difference between areas.

EXERCISE 5

1. $25.32 \text{ m}^2 + 62.28 \text{ m}^2 = \underline{\hspace{2cm}}$
2. $6.75 \text{ ha} - 4.25 \text{ ha} = \underline{\hspace{2cm}}$
3. A garden is 30 m² and another is 45 m². Find their total area.
4. Two fields are 5 ha and 3 ha. Find the difference.

7. REAL-LIFE PROBLEMS

Area and land measurements are important in:

- Farming
- Buying and selling land
- Construction
- Planning gardens and playgrounds

Example:

1 a of land produces 2 tonnes of oranges.

$$300 \text{ m}^2 = 3 \text{ a}$$

$$\text{Production} = 3 \times 2 = 6 \text{ tonnes}$$

$$6 \text{ tonnes} = 6,000 \text{ kg.}$$

EXERCISE 6

1. Kenia has land measuring $100 \text{ m} \times 100 \text{ m}$. Find its area in hectares.
2. Peter has 3 ha and Mary has 200 a. Who has more land?
3. Find their total land in hectares.

UNIT 9 REVISION EXERCISE

1. Convert:

- a) $2,500 \text{ cm}^2$ to m^2
- b) $8,000 \text{ m}^2$ to hm^2
- c) 6.5 km^2 to dam^2
- d) 5 m^2 to cm^2
- e) 10 ha to m^2

2. A paper has area 6 cm^2 and length 3 cm. Find its width.

3. Calculate:

- a) $25.32 \text{ m}^2 + 62.28 \text{ m}^2$
- b) $6.75 \text{ ha} - 4.25 \text{ ha}$

4. Find the area:

- a) $50 \text{ m} \times 20 \text{ m}$
- b) $100 \text{ m} \times 50 \text{ m}$

5. A classroom floor is 8 m long and 10 m wide. Find its area.

6. Lake Kivu has an area of $2,700 \text{ km}^2$ and Lake Burera 55 km^2 .
By how much is Lake Kivu bigger?

7. A farmer shares 2 ha equally among 4 children.
Find each child's share in m^2 .

QUICK REVISION

- Area = space inside a shape.
- Standard unit of area = m^2 .
- Rectangle: Area = Length $\times$ Width.
- Square: Area = Side $\times$ Side.
- Area units change by $\times 100$ or $\div 100$ per step.
- $1 \text{ ha} = 100 \text{ a} = 10,000 \text{ m}^2$.
- $1 \text{ a} = 100 \text{ m}^2$.

- $1 \text{ ca} = 1 \text{ m}^2$.
- $1 \text{ hm}^2 = 1 \text{ ha}$.
- $1 \text{ dam}^2 = 1 \text{ a}$.
- Convert to the same unit before adding, subtracting or comparing areas.

END OF UNIT 9

UNIT 10: TIME

1. MEANING OF TIME

Time helps us know when an activity happens and how long it takes.

We use clocks and watches to tell time and calendars to identify days, dates, weeks and months.

2. READING AND TELLING TIME

An analogue clock has:

- Hour hand – the short hand that shows hours.
- Minute hand – the long hand that shows minutes.
- Second hand – shows seconds.

Important facts:

- 1 complete turn of the hour hand = 12 hours.
- 1 complete turn of the minute hand = 60 minutes.
- 1 complete turn of the second hand = 60 seconds.
- From one number to the next on a clock = 5 minutes.

Examples:

6:00 = six o'clock

6:15 = quarter past six

6:30 = half past six

6:45 = quarter to seven

7:20 = twenty minutes past seven

7:50 = ten minutes to eight

Remember:

Quarter past = 15 minutes after the hour.

Half past = 30 minutes after the hour.

Quarter to = 15 minutes before the next hour.

3. A.M. AND P.M.

A.M. is used for time from midnight to before midday.

P.M. is used for time from midday to before midnight.

Examples:

6:00 in the morning = 6:00 a.m.

10:30 in the morning = 10:30 a.m.

2:00 in the afternoon = 2:00 p.m.

7:00 in the evening = 7:00 p.m.

12:00 during the day = noon.

12:00 at night = midnight.

4. UNITS OF TIME

The standard unit of time is the second.

Important conversions:

1 minute = 60 seconds

1 hour = 60 minutes

1 day = 24 hours

1 week = 7 days

Other units include:

- Month
- Year
- Decade
- Century
- Millennium

Examples:

3 hours = $3 \times 60 = 180$ minutes.

120 minutes = $120 \div 60 = 2$ hours.

2 days = $2 \times 24 = 48$ hours.

3 weeks = $3 \times 7 = 21$ days.

5. READING A CALENDAR

A calendar shows days, dates, weeks and months of a year.

There are 7 days in a week:

Monday, Tuesday, Wednesday, Thursday, Friday, Saturday and Sunday.

There are 12 months in a year:

January, February, March, April, May, June, July, August, September, October, November and December.

Months with 31 days:

January, March, May, July, August, October and December.

Months with 30 days:

April, June, September and November.

February normally has 28 days.

6. FINDING DURATION

Duration means the amount of time an activity takes.

Duration = Finishing time - Starting time

Example:

A lesson starts at 8:00 a.m. and ends at 8:40 a.m.

Duration = 40 minutes.

Example:

A football match starts at 3:00 p.m. and ends at 5:00 p.m.

Duration = 2 hours.

7. FINDING FINISHING TIME

To find the finishing time, add the duration to the starting time.

Example:

A lesson starts at 10:00 a.m. and lasts 40 minutes.

Finishing time = 10:40 a.m.

Example:

A journey starts at 2:00 p.m. and lasts 3 hours.

Finishing time = 5:00 p.m.

8. FINDING DAYS AND DATES

We can count forward or backward on a calendar.

Example:

Today is Monday.

Three days later:

Tuesday → Wednesday → Thursday

Therefore, three days after Monday is Thursday.

Remember:

1 week = 7 days

2 weeks = 14 days

3 weeks = 21 days

4 weeks = 28 days

9. AVERAGE TIME

Average time is found by dividing the total time by the number of activities or items.

Average time = Total time ÷ Number of items

Example:

Three activities take 2 hours, 3 hours and 4 hours.

Total = $2 + 3 + 4 = 9$ hours

Average = $9 \div 3 = 3$ hours.

10. IMPORTANCE OF TIME

Time helps us:

- Arrive at school on time.
- Plan lessons and activities.
- Know when events start and finish.
- Plan journeys.
- Follow daily schedules.
- Know dates and special events.

SHORT REVISION EXERCISE

1. Convert:

- a) 3 hours to minutes.
- b) 120 seconds to minutes.
- c) 2 days to hours.
- d) 4 weeks to days.

2. Write using a.m. or p.m.:

- a) 7 o'clock in the morning.
- b) 4 o'clock in the afternoon.

3. A lesson starts at 9:00 a.m. and ends at 9:40 a.m.

How long is the lesson?

4. Today is Tuesday. What day will it be after 5 days?

QUICK REVISION

- 1 minute = 60 seconds.
- 1 hour = 60 minutes.
- 1 day = 24 hours.
- 1 week = 7 days.
- A year has 12 months.
- Quarter past = 15 minutes after.
- Half past = 30 minutes after.
- Quarter to = 15 minutes before the next hour.
- A.M. = before midday.
- P.M. = after midday.
- Duration = finishing time - starting time.
- A calendar shows days, dates, weeks and months.

END OF UNIT 10

UNIT 11: NUMBER PATTERNS

1. MEANING OF A NUMBER PATTERN

A number pattern is a sequence of numbers that follows a particular rule.

Examples:

2, 4, 6, 8, 10, ...

Rule: Add 2 each time.

20, 18, 16, 14, 12, ...

Rule: Subtract 2 each time.

Number patterns help us find missing numbers and predict the numbers that come next.

2. ASCENDING AND DESCENDING ORDER

Ascending order means arranging numbers from the smallest to the largest.

Example:

900, 9, 90, 9,000

Ascending order:

9, 90, 900, 9,000

Descending order means arranging numbers from the largest to the smallest.

Descending order:

9,000, 900, 90, 9

Remember:

Ascending = Smallest to largest.

Descending = Largest to smallest.

3. ARITHMETIC PROGRESSION

An arithmetic progression is a number pattern in which the same number is added or subtracted each time.

Example 1:

2, 5, 8, 11, 14, ...

Rule: Add 3.

Example 2:

20, 17, 14, 11, 8, ...

Rule: Subtract 3.

To identify the rule, compare one number with the next.

Example:

10, 20, 30, 40, 50, ...

Rule: Add 10.

4. FINDING MISSING NUMBERS

First identify the rule, then use it to find the missing number.

Example:

90, 80, ____, 60, 50

Rule: Subtract 10.

$$80 - 10 = 70$$

Missing number = 70.

Another example:

2, 4, 6, ____, 10

Rule: Add 2.

Missing number = 8.

5. GEOMETRIC PROGRESSION

A geometric progression is a number pattern in which numbers are repeatedly multiplied or divided by the same number.

Example:

2, 4, 8, 16, 32, ...

Rule: Multiply by 2.

Another example:

64, 32, 16, 8, 4, ...

Rule: Divide by 2.

6. FINDING MISSING NUMBERS IN GEOMETRIC PATTERNS

Example:

3, 6, 12, 24, ____

Rule: Multiply by 2.

$$24 \times 2 = 48$$

Missing number = 48.

Example:

100, 50, 25, ____

Rule: Divide by 2.

$$25 \div 2 = 12.5$$

Missing number = 12.5.

7. DIFFERENCE BETWEEN ARITHMETIC AND GEOMETRIC PATTERNS

Arithmetic progression:

- Uses repeated addition or subtraction.

Example:

5, 10, 15, 20, 25

Rule: Add 5.

Geometric progression:

- Uses repeated multiplication or division.

Example:

2, 4, 8, 16, 32

Rule: Multiply by 2.

8. IMPORTANCE OF NUMBER PATTERNS

Number patterns can be used in:

- Counting.
- Saving money.
- Planning activities.
- Finding missing values.
- Predicting what comes next.
- Solving mathematical problems.

SHORT REVISION EXERCISE

1. Find the next two numbers:

a) 5, 10, 15, 20, ____, ____

b) 30, 25, 20, 15, ____, ____

2. Find the missing number:

a) 4, 8, 12, ____, 20

b) 2, 4, 8, 16, ____

3. State whether each pattern is arithmetic or geometric:

a) 3, 6, 9, 12, 15

b) 5, 10, 20, 40, 80

QUICK REVISION

- Number pattern = numbers following a rule.
- Ascending = smallest to largest.
- Descending = largest to smallest.
- Arithmetic progression uses addition or subtraction.
- Geometric progression uses multiplication or division.
- Always find the rule before finding a missing number.

END OF UNIT 11

UNIT 12: SOLVING PROBLEMS INVOLVING MISSING NUMBERS IN ADDITION AND SUBTRACTION

1. MEANING OF A MISSING NUMBER

A missing number is a number that is not shown in a mathematical sentence.

Example:

$$\underline{\quad} + 5 = 12$$

To find the missing number:

$$12 - 5 = 7$$

Therefore:

$$7 + 5 = 12.$$

Finding missing numbers helps us solve puzzles and everyday problems involving money, sharing and counting.

2. MISSING NUMBERS IN ADDITION

When one number in an addition sentence is missing, subtract the known number from the total.

Example:

$$5 + \underline{\quad} = 8$$

Missing number:

$$8 - 5 = 3$$

Therefore:

$$5 + 3 = 8.$$

Another example:

$$\underline{\quad} + 9 = 13$$

$$13 - 9 = 4$$

Missing number = 4.

RULE:

Missing addend = Total - Known addend

3. ARITHMAGONS

An arithmagon is a mathematical puzzle where numbers are placed at the vertices of a shape.

Numbers between the vertices can be found by adding or subtracting the vertex numbers.

Example:

If the vertex numbers are 2, 7, 3 and 12:

$$2 + 7 = 9$$

$$2 + 12 = 14$$

$$12 + 3 = 15$$

$$7 + 3 = 10$$

Arithmagons help learners practise finding missing numbers.

:contentReference[oaicite:1]{index=1}

4. MISSING NUMBERS IN SUBTRACTION

CASE 1:

The number being subtracted is missing.

Example:

$$20 - \underline{\quad} = 12$$

$$20 - 12 = 8$$

Missing number = 8.

CASE 2:

The first number is missing.

Example:

$$\underline{\quad} - 5 = 10$$

$$10 + 5 = 15$$

Missing number = 15.

Remember:

$$20 - \underline{\quad} = 12$$

Use subtraction.

$$\underline{\quad} - 5 = 10$$

Use addition. :contentReference[oaicite:2]{index=2}

5. USING INVERSE OPERATIONS

Addition and subtraction are inverse operations.

This means one operation can undo the other.

Examples:

$$8 + 5 = 13$$

$$13 - 5 = 8$$

$$15 - 7 = 8$$

$$8 + 7 = 15$$

Inverse operations help us find and check missing numbers.

6. PROBLEMS WITH ADDITION AND SUBTRACTION TOGETHER

Some problems use both operations.

Example:

$$25 - \underline{\quad} + 5 = 22$$

First combine:

$$25 + 5 = 30$$

Now:

$$30 - \underline{\quad} = 22$$

Missing number:

$$30 - 22 = 8$$

Therefore:

$$25 - 8 + 5 = 22. :contentReference[oaicite:3]{index=3}$$

7. SOLVING WORD PROBLEMS

Read the problem carefully and decide whether to add or subtract.

Example:

Lilly had 1,000 Frw.

She bought a notebook for 400 Frw and a toy for 250 Frw.

Total money spent:

$$400 + 250 = 650 \text{ Frw}$$

Money remaining:

$$1,000 - 650 = 350 \text{ Frw}$$

Therefore:

She spent 650 Frw and remained with 350 Frw. :contentReference[oaicite:4]{index=4}

IMPORTANT RULES

For:

$$\underline{\quad} + 5 = 12$$

Use:

$$12 - 5$$

For:

$$5 + \underline{\quad} = 12$$

Use:

$$12 - 5$$

For:

$$\underline{\quad} - 5 = 10$$

Use:

$$10 + 5$$

For:

$$20 - \underline{\quad} = 12$$

Use:

$$20 - 12$$

Always check the answer by putting it back into the original number sentence.

SHORT REVISION EXERCISE

1. Find the missing number:

a) $500 + \underline{\quad} = 805$

b) $\underline{\quad} - 150 = 450$

c) $14 - \underline{\quad} = 8$

d) $\underline{\quad} - 8 = 19$

2. A school has 643 pupils. There are 372 boys. How many girls are there?

3. John had 30 apples, gave away 12 and later bought 15 more. How many apples does he have now?

QUICK REVISION

- A missing number is an unknown number in a calculation.
- Use subtraction to find a missing number in addition.
- Addition and subtraction are inverse operations.
- In $\underline{\quad} - 5 = 10$, add $10 + 5$.
- In $20 - \underline{\quad} = 12$, subtract $20 - 12$.
- An arithmagon is a number puzzle.

- Always check your answer using the original calculation.

END OF UNIT 12

UNIT 13: LINES AND ANGLES

1. MEANING AND IMPORTANCE OF LINES

Lines are found everywhere. They are important in drawing, designing and making objects such as tables, doors and windows.

Common types of lines:

- Vertical line – goes up and down, like a flagpole.
- Horizontal line – goes from side to side, like the top of a table.
- Oblique line – slants; it is neither vertical nor horizontal.
- Curved line – bends smoothly.
- Intersecting lines – two lines that cross at a point.
- Perpendicular lines – meet at a right angle of 90° .
- Parallel lines – move in the same direction and never meet.

2. LINE SEGMENT

A line segment is a straight line with two endpoints.

Example:

A ----- B

This is line segment AB.

A line segment starts at one point and ends at another.

A line is different because it continues in both directions.

A ruler is used to draw and measure line segments.

Example:

A line segment AB can be 3 cm long.

3. MEANING OF AN ANGLE

An angle is formed when two lines meet at a point.

An angle measures the space between two intersecting lines.

When two straight lines cross, they can form four angles.

Angles are found in objects such as doors, scissors, tables and buildings.

4. MEASURING ANGLES

Angles are measured in degrees.

The symbol for degrees is $^{\circ}$.

Example:

60 degrees = 60°

A protractor is used to measure angles.

A protractor can measure angles from 0° to 180° .

Steps for measuring an angle:

1. Place the centre of the protractor at the point where the lines meet.
2. Align one line with the horizontal line of the protractor.
3. Read the number where the other line crosses the protractor.

Example:

If the protractor shows 70° , the angle measures 70 degrees.

5. TYPES OF ANGLES

A. RIGHT ANGLE

A right angle measures exactly 90° .

B. ACUTE ANGLE

An acute angle is greater than 0° but less than 90° .

Example:

30° is an acute angle.

C. OBTUSE ANGLE

An obtuse angle is greater than 90° but less than 180° .

Example:

135° is an obtuse angle.

D. STRAIGHT ANGLE

A straight angle measures exactly 180° .

E. REFLEX ANGLE

A reflex angle is greater than 180° but less than 360° .

Examples:

285° and 339° are reflex angles.

6. COMPLEMENTARY ANGLES

Complementary angles are two angles whose total is 90° .

Example:

$$27^\circ + 63^\circ = 90^\circ$$

Therefore, 27° and 63° are complementary angles.

To find a missing complementary angle:

$$90^\circ - \text{Given angle}$$

Example:

Find the complement of 40° .

$$90^\circ - 40^\circ = 50^\circ$$

Answer = 50° .

7. SUPPLEMENTARY ANGLES

Supplementary angles are two angles whose total is 180° .

Example:

$$121^\circ + 59^\circ = 180^\circ$$

Therefore, 121° and 59° are supplementary angles.

To find a missing supplementary angle:

$$180^\circ - \text{Given angle}$$

Example:

Find the supplement of 120° .

$$180^\circ - 120^\circ = 60^\circ$$

Answer = 60° .

8. USES OF LINES AND ANGLES

Lines and angles are useful in:

- Drawing.
- Designing.
- Building houses.
- Making tables and furniture.
- Making doors and windows.
- Construction and other geometric work.

Accuracy is important when drawing lines and measuring angles.

SHORT REVISION EXERCISE

1. Name each angle:

- a) 30°
- b) 90°
- c) 120°
- d) 180°

2. Find:

- a) Complement of 30° .
- b) Supplement of 120° .

3. Draw:

- a) Parallel lines.
- b) Perpendicular lines.
- c) A line segment AB of 5 cm.
- d) An angle of 60° using a protractor.

QUICK REVISION

- Vertical line = up and down.
- Horizontal line = side to side.
- Parallel lines never meet.
- Perpendicular lines meet at 90° .
- Intersecting lines cross each other.
- A line segment has two endpoints.
- Angles are measured in degrees.
- A protractor measures angles.
- Acute angle = less than 90° .
- Right angle = exactly 90° .
- Obtuse angle = between 90° and 180° .
- Straight angle = exactly 180° .
- Reflex angle = between 180° and 360° .
- Complementary angles add up to 90° .
- Supplementary angles add up to 180° .

END OF UNIT 13

UNIT 14: SHAPES IN THE PLANE

1. MEANING OF SHAPES IN THE PLANE

Shapes in the plane are flat or two-dimensional (2D) shapes.

They are useful in everyday life.

Examples:

- Wheels, clocks and coins have circular shapes.
- Houses, books and tables use squares and rectangles.
- Bridges and roofs may use triangular shapes.

Shapes can be classified using their sides, angles, diagonals and lines of symmetry.

2. NAMING TRIANGLES ACCORDING TO THEIR SIDES

A triangle has three sides.

Triangles can be named according to the lengths of their sides.

EQUILATERAL TRIANGLE

- All three sides are equal.

ISOSCELES TRIANGLE

- Two sides are equal.
- One side is different.

SCALENE TRIANGLE

- All three sides have different lengths.

3. NAMING TRIANGLES ACCORDING TO THEIR ANGLES

A triangle has:

- 3 sides
- 3 angles
- 3 vertices

A vertex is a point where two sides meet.

Triangles can also be named according to their angles.

ACUTE TRIANGLE

- All its angles are less than 90° .

RIGHT-ANGLED TRIANGLE

- One angle is exactly 90° .

OBTUSE TRIANGLE

- One angle is greater than 90° .

4. TRIANGLES AND QUADRILATERALS

A triangle has:

- 3 sides
- 3 angles

A quadrilateral has:

- 4 sides
- 4 angles
- 4 vertices

Examples of quadrilaterals:

- Square

- Rectangle
- Rhombus
- Parallelogram
- Trapezium

We can distinguish these shapes by looking at their sides, angles, diagonals and symmetry.

5. SQUARE

A square is a quadrilateral.

Properties:

- All four sides are equal.
- All four angles are 90° .
- Opposite sides are parallel.
- Diagonals are equal.
- Diagonals meet in the middle.
- Diagonals meet at right angles.
- It has 4 lines of symmetry.
- Order of rotational symmetry = 4.

6. RECTANGLE

Properties:

- Opposite sides are equal.
- Opposite sides are parallel.
- All four angles are 90° .
- Diagonals are equal.
- Diagonals meet in the middle.
- Diagonals do not meet at right angles.
- It has 2 lines of symmetry.
- Order of rotational symmetry = 2.

7. RHOMBUS

Properties:

- All four sides are equal.
- Two angles are acute.
- Two angles are obtuse.
- Diagonals meet in the middle.
- Diagonals meet at right angles.
- It has 2 lines of symmetry.
- Order of rotational symmetry = 2.

8. PARALLELOGRAM

Properties:

- Opposite sides are equal.
- Opposite sides are parallel.
- Diagonals are not equal.
- Diagonals meet in the middle.

- Diagonals do not meet at right angles.
- It has no line of symmetry.
- Order of rotational symmetry = 2.

9. TRAPEZIUM

Properties:

- One pair of opposite sides is parallel.
- The other two sides are not parallel.
- Diagonals do not meet in the middle.
- Diagonals do not meet at right angles.
- It usually has no line of symmetry.

An isosceles trapezium has 1 line of symmetry.

10. DIAGONALS

A diagonal joins one vertex of a shape to another vertex that is not next to it.

Important properties:

Square:

- Diagonals are equal.
- They meet in the middle at right angles.

Rectangle:

- Diagonals are equal.
- They meet in the middle but not at right angles.

Rhombus:

- Diagonals meet in the middle at right angles.

Parallelogram:

- Diagonals are not equal.
- They meet in the middle but not at right angles.

Trapezium:

- Diagonals do not meet in the middle.

11. LINES OF SYMMETRY

A line of symmetry divides a shape into two equal matching parts.

If the shape is folded along the line of symmetry, the two parts match.

Number of lines of symmetry:

Square = 4

Rectangle = 2

Rhombus = 2

Parallelogram = 0

Trapezium = usually 0

Isosceles trapezium = 1

12. COMPARING QUADRILATERALS

SQUARE

- All sides equal.
- All angles are 90° .

RECTANGLE

- Opposite sides equal.
- All angles are 90° .

RHOMBUS

- All sides equal.
- Has acute and obtuse angles.

PARALLELOGRAM

- Opposite sides equal and parallel.
- Diagonals meet in the middle.

TRAPEZIUM

- One pair of opposite sides is parallel.

13. IMPORTANCE OF SHAPES

Shapes are important in:

- Building houses.
- Designing furniture.
- Making roads and bridges.
- Drawing.
- Making books and tables.
- Designing everyday objects.

Builders and designers need to understand the properties of shapes so that their work is accurate.

SHORT REVISION EXERCISE

1. Name the triangle:

- a) All three sides are equal.
- b) Two sides are equal.
- c) All three sides are different.

2. Name the triangle:

- a) All angles are less than 90° .
- b) One angle is 90° .

c) One angle is greater than 90° .

3. How many lines of symmetry does each shape have?

- a) Square
- b) Rectangle
- c) Rhombus
- d) Parallelogram

4. Name the quadrilateral:

- a) Four equal sides and four right angles.
- b) Opposite sides equal and four right angles.
- c) Four equal sides with acute and obtuse angles.

QUICK REVISION

- Triangle = 3 sides and 3 angles.
- Equilateral triangle = 3 equal sides.
- Isosceles triangle = 2 equal sides.
- Scalene triangle = all sides different.
- Acute triangle = all angles less than 90° .
- Right-angled triangle = one angle is 90° .
- Obtuse triangle = one angle is greater than 90° .
- Quadrilateral = 4 sides and 4 angles.
- Square = 4 equal sides and 4 right angles.
- Rectangle = opposite sides equal and 4 right angles.
- Rhombus = 4 equal sides.
- Parallelogram = opposite sides equal and parallel.
- Trapezium = one pair of opposite sides parallel.
- Line of symmetry divides a shape into two equal matching parts.

END OF UNIT 14

UNIT 15: AREA OF SHAPES IN THE PLANE

1. MEANING OF AREA

Area tells us how much space a flat shape covers.

Area is useful in everyday life. For example:

- Farmers use area to measure land for planting crops.
- Engineers use area when planning buildings.
- Tailors use area when measuring material for clothes.

Area is measured in square units such as:

- Square centimetres = cm^2
- Square metres = m^2

Area is different from perimeter:

- Area = space inside a shape.
- Perimeter = distance around a shape.

2. AREA OF A RECTANGLE

A rectangle has opposite sides that are equal.

Formula:

$$\text{Area} = \text{Length} \times \text{Width}$$

Example:

$$\text{Length} = 6 \text{ m}$$

$$\text{Width} = 4 \text{ m}$$

$$\text{Area} = 6 \times 4$$

$$\text{Area} = 24 \text{ m}^2$$

Therefore, the rectangle covers 24 square metres.

3. AREA OF A SQUARE

A square has four equal sides.

Formula:

$$\text{Area} = \text{Side} \times \text{Side}$$

Example:

$$\text{Side} = 4 \text{ m}$$

$$\text{Area} = 4 \times 4$$

$$\text{Area} = 16 \text{ m}^2$$

If the area is given, we can also find the side.

Example:

$$\text{Area} = 81 \text{ m}^2$$

$$9 \text{ m} \times 9 \text{ m} = 81 \text{ m}^2$$

Therefore:

$$\text{Side} = 9 \text{ m}.$$

4. AREA OF A TRIANGLE

A triangle has three sides.

Formula:

$$\text{Area} = \text{Base} \times \text{Height} \div 2$$

The base and height must meet at a right angle.

Example:

Base = 12 m
Height = 7 m

Area = $12 \times 7 \div 2$
Area = $84 \div 2$
Area = 42 m^2

5. AREA OF A PARALLELOGRAM

A parallelogram has four sides. Its opposite sides are equal and parallel.

Formula:
Area = Base $\times$ Height

The height is a straight line from the base to the top that meets the base at 90° .

Example:
Base = 8 cm
Height = 4 cm

Area = 8×4
Area = 32 cm^2

6. AREA OF A RHOMBUS

A rhombus is a four-sided shape with all four sides equal.

It has two diagonals.

Formula:
Area = Diagonal 1 $\times$ Diagonal 2 $\div$ 2

Example:
Diagonal 1 = 20 cm
Diagonal 2 = 32 cm

Area = $20 \times 32 \div 2$
Area = $640 \div 2$
Area = 320 cm^2

The two diagonals divide a rhombus into four equal right-angled triangles.

7. AREA OF A TRAPEZIUM

A trapezium is a four-sided shape with one pair of opposite sides parallel.

Formula:
Area = (Base 1 + Base 2) $\times$ Height $\div$ 2

Base 1 and Base 2 are the parallel sides.

The height is perpendicular to the parallel sides.

Example:

Base 1 = 23 cm

Base 2 = 17 cm

Height = 15 cm

$$\text{Area} = (23 + 17) \times 15 \div 2$$

$$\text{Area} = 40 \times 15 \div 2$$

$$\text{Area} = 600 \div 2$$

$$\text{Area} = 300 \text{ cm}^2$$

A diagonal can divide a trapezium into two triangles. The area of the trapezium can therefore be found by adding the areas of the two triangles.

8. FORMULAS TO REMEMBER

Rectangle:

$$\text{Area} = \text{Length} \times \text{Width}$$

Square:

$$\text{Area} = \text{Side} \times \text{Side}$$

Triangle:

$$\text{Area} = \text{Base} \times \text{Height} \div 2$$

Parallelogram:

$$\text{Area} = \text{Base} \times \text{Height}$$

Rhombus:

$$\text{Area} = \text{Diagonal 1} \times \text{Diagonal 2} \div 2$$

Trapezium:

$$\text{Area} = (\text{Base 1} + \text{Base 2}) \times \text{Height} \div 2$$

9. IMPORTANCE OF ACCURATE MEASUREMENT

Correct measurements are important when finding area.

Accurate measurement helps:

- Builders calculate the correct size of roofs and floors.
- Farmers know the size of their land.
- Tailors cut the correct amount of cloth.
- People avoid wasting materials.

SHORT REVISION EXERCISE

1. Find the area of a rectangle measuring 8 m by 5 m.
2. Find the area of a square with a side of 9 cm.
3. Find the area of a triangle with:
Base = 10 m
Height = 6 m
4. Find the area of a parallelogram with:
Base = 12 cm
Height = 5 cm
5. A rhombus has diagonals of 20 m and 10 m.
Find its area.
6. A trapezium has parallel sides of 7 m and 11 m and a height of 6 m.
Find its area.

QUICK REVISION

- Area = space covered by a flat shape.
- Area is written in square units.
- Rectangle: $\text{Area} = \text{Length} \times \text{Width}$.
- Square: $\text{Area} = \text{Side} \times \text{Side}$.
- Triangle: $\text{Area} = \text{Base} \times \text{Height} \div 2$.
- Parallelogram: $\text{Area} = \text{Base} \times \text{Height}$.
- Rhombus: $\text{Area} = \text{Diagonal 1} \times \text{Diagonal 2} \div 2$.
- Trapezium: $\text{Area} = (\text{Base 1} + \text{Base 2}) \times \text{Height} \div 2$.
- Area measures inside a shape.
- Perimeter measures around a shape.
- Always use the correct unit in the final answer.

END OF UNIT 15

UNIT 16: INTRODUCTION TO DATA HANDLING

1. MEANING OF DATA AND STATISTICS

Data is information that we collect about people, objects, events or other things.

Statistics helps us to:

- Collect data.
- Organize data.
- Present data.
- Understand and interpret data.
- Make good decisions.

Examples:

- Teachers use statistics to study pupils' marks.

- Shopkeepers use it to know how many items they sell.
- Doctors can use it to study illnesses in a community.

2. TYPES OF DATA

There are two main types of data:

A. QUALITATIVE DATA

Qualitative data describes things using words.

It tells us about qualities such as:

- Colour
- Taste
- Size
- Type
- Appearance

Examples:

- The apple is red.
- The blanket is soft.
- The cake is sweet.
- My favourite colour is blue.

B. QUANTITATIVE DATA

Quantitative data uses numbers to count or measure things.

Examples:

- There are 5 apples.
- The rope is 2 metres long.
- The cake weighs 1 kilogram.
- A child is 10 years old.

Remember:

Qualitative = descriptions or words.

Quantitative = numbers or measurements.

3. WAYS OF COLLECTING DATA

There are three main ways of collecting data:

A. OBSERVATION

Observation means watching and recording what you see.

Example:

Counting the number of birds in the school compound.

B. SURVEY AND INTERVIEW

This involves asking people questions and recording their answers.

Example:

Asking classmates about their favourite fruits.

C. EXPERIMENT

An experiment involves testing something and recording the results.

Example:

Measuring how long different pupils take to run across a playground.

4. PRESENTATION OF DATA

Presentation of data means displaying collected information in a simple and understandable way.

Data can be presented using:

- Tables
- Bar graphs
- Pie charts
- Line graphs
- Pictographs

In this unit, the main focus is on tables and bar graphs.

5. PRESENTING DATA USING TABLES

A table organizes information in rows and columns.

Example:

Fruit	Number of pupils
Apple	5
Banana	8
Orange	4
Mango	6

Tables make information easy to read and compare.

The number of times something appears in data is called its frequency.

6. PRESENTING DATA USING BAR GRAPHS

A bar graph uses bars to represent and compare numbers.

A bar graph has two main axes:

- X-axis = horizontal axis.
- Y-axis = vertical axis.

Usually:

- The X-axis shows categories such as names, classes or fruits.

- The Y-axis shows numbers or quantities.

The height of each bar represents the value.

Remember:

The taller the bar, the bigger the number.

Example:

If 3 pupils like apples and 8 pupils like bananas, the banana bar will be taller.

7. STEPS FOR DRAWING A BAR GRAPH

1. Read the data carefully.
2. Draw the horizontal X-axis.
3. Draw the vertical Y-axis.
4. Write the categories on the X-axis.
5. Write the numbers on the Y-axis using a suitable scale.
6. Draw bars to represent the values.
7. Give the graph a suitable title.

Bars should be easy to read and compare.

8. INTERPRETING TABLES

Interpreting a table means reading and understanding the information shown in it.

From a table, we can find:

- The highest value.
- The lowest value.
- The total.
- Equal values.
- Differences between values.
- Most and least common items.

Example:

Day	Rainfall
Monday	10 mm
Tuesday	20 mm
Wednesday	40 mm
Thursday	15 mm
Friday	20 mm

From the table:

- Highest rainfall = 40 mm on Wednesday.
- Lowest rainfall = 10 mm on Monday.
- Tuesday and Friday have equal rainfall.
- Difference between Wednesday and Monday:
 $40 - 10 = 30$ mm.

9. INTERPRETING BAR GRAPHS

Interpreting a bar graph means reading the bars and using them to answer questions or make conclusions.

Steps:

1. Read the title.
2. Look at what each bar represents.
3. Check the numbers on the Y-axis.
4. Compare the heights of the bars.
5. Identify the tallest and shortest bars.
6. Use the values to answer questions.

Example:

If a graph shows:

Apple = 5 pupils

Banana = 3 pupils

Grapes = 8 pupils

We can conclude:

- Grapes are the most popular.
- Bananas are the least popular.
- 5 pupils like apples.

10. IMPORTANCE OF DATA HANDLING

Data handling helps people:

- Organize information.
- Compare information.
- Make decisions.
- Identify patterns.
- Understand results.
- Solve everyday problems.

It is important to collect and record data correctly because incorrect data can lead to wrong decisions.

SHORT REVISION EXERCISE

1. State whether each is qualitative or quantitative:

- a) Favourite colour.
- b) Number of books.
- c) Age of a pupil.
- d) Type of weather.

2. Name three ways of collecting data.

3. What is the difference between the X-axis and Y-axis of a bar graph?

4. Study the table:

Fruit	Number of pupils
Apple	10
Banana	15
Orange	8
Mango	12

- a) Which fruit is most popular?
- b) Which fruit is least popular?
- c) Find the total number of pupils.

QUICK REVISION

- Data = collected information.
- Statistics helps us collect, organize, present and understand data.
- Qualitative data uses descriptions or words.
- Quantitative data uses numbers or measurements.
- Data can be collected through observation, surveys/interviews and experiments.
- A table organizes data in rows and columns.
- Frequency tells how many times something occurs.
- A bar graph uses bars to represent data.
- X-axis = horizontal axis.
- Y-axis = vertical axis.
- Taller bar = greater value.
- Interpreting data means reading and understanding the information.
- Correct data helps us make good decisions.

END OF UNIT 16

UNIT 17: INTRODUCTION TO GAMES OF CHANCE

1. MEANING OF PROBABILITY

Probability helps us predict what might happen.

An event can be:

- Certain – it will happen.
- Possible – it might happen.
- Impossible – it cannot happen.

Example:

If the sky is dark and cloudy, it may rain.

Games of chance are games where we cannot know the exact result before playing.

Examples include:

- Tossing a coin.
- Throwing a die.

- Playing cards.

2. TOSSING A COIN

A coin has two sides:

- Head (H)
- Tail (T)

When a coin is tossed, either head or tail can face up.

Each side has an equal chance of facing up.

Important:

- Sometimes heads may appear more often.
- Sometimes tails may appear more often.
- We cannot know for sure which side will appear before tossing.
- Tossing a coin is therefore a game of chance.

Example:

A coin is tossed 3 times.

Possible result:

1st toss = Head

2nd toss = Tail

3rd toss = Head

If Head is worth 1 mark and Tail is worth 0 marks:

Total marks = $1 + 0 + 1 = 2$ marks.

3. TOSSING THREE COINS AT THE SAME TIME

Three coins can be tossed together.

Possible results include:

- HHH – three heads
- TTT – three tails
- HHT – two heads and one tail
- HTH – two heads and one tail
- TTH – two tails and one head

Each coin has an equal chance of showing head or tail.

The results may change each time the game is played.

A winning combination should be decided before the game begins.

Example:

Winning combination = two heads and one tail.

If the result is HHT, the player wins.

4. PLAYING CARDS

A standard pack has 52 playing cards.

The cards are divided into 4 suits:

- Hearts
- Diamonds
- Clubs
- Spades

Each suit has 13 cards:

A, 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q and K.

Playing cards have two colours:

- Red – Hearts and Diamonds.
- Black – Clubs and Spades.

Card games can be used to learn about chance.

Example:

A player may try to guess:

- The colour of a card.
- The suit of a card.
- The number or face on the card.

The result cannot be known for sure before the card is picked.

5. THROWING A DIE

A die has 6 sides.

The sides are numbered:

1, 2, 3, 4, 5 and 6.

When a die is thrown, one number faces up.

Each number has an equal chance of appearing.

We may guess the result, but we cannot know exactly which number will appear.

Example:

A die is thrown four times and gives:

2, 5, 3, 3

Total score:

$$2 + 5 + 3 + 3 = 13.$$

6. FAIR AND UNFAIR GAMES

A fair game does not favour one player over another.

All players should:

- Follow the same rules.
- Have a fair chance of winning.
- Take turns properly.
- Play honestly.

A game may be unfair if:

- One player is given an advantage.
- Players do not follow the same rules.
- Someone cheats.
- The rules favour one player.

Example:

If two players throw the same die the same number of times under the same rules, the game can be fair.

7. IMPORTANCE OF RULES IN GAMES OF CHANCE

Rules help make games fair and enjoyable.

Players should:

- Follow the rules.
- Respect other players.
- Wait for their turn.
- Record results honestly.
- Avoid cheating.
- Accept the result of a fair game.

It is important to speak against unfair games instead of ignoring unfair behaviour.

8. IMPORTANCE OF LEARNING GAMES OF CHANCE

Games of chance help us:

- Understand probability.
- Make predictions.
- Understand uncertain events.
- Practise recording results.
- Learn about fairness.
- Make better decisions.

Probability can also help in areas such as weather forecasting and assessing risks.

SHORT REVISION EXERCISE

1. How many sides does a coin have?
2. How many sides does a die have?
3. How many cards are in a standard pack of playing cards?
4. Name the four suits of playing cards.
5. Give two possible results when three coins are tossed together.
6. Explain the difference between a fair game and an unfair game.

QUICK REVISION

- Probability helps us predict what might happen.
- A game of chance has an uncertain result.
- A coin has 2 sides: Head and Tail.
- Head and Tail have equal chances of facing up.
- Three coins can produce combinations such as HHH, TTT and HHT.
- A standard pack contains 52 cards.
- There are 4 suits: Hearts, Diamonds, Clubs and Spades.
- Each suit contains 13 cards.
- A die has 6 sides numbered 1 to 6.
- Each number on a die has an equal chance of appearing.
- A fair game does not favour one player.
- Players should follow rules, take turns and avoid cheating.

END OF UNIT 17